

Brief History of Visual Aids

Cave Paintings

The 2,000 images found in the caves at Lascaux, France narrate stories through character, sequence, and motion. The oldest evidence the world has of visual storytelling, the paintings demonstrate early reliance on using images to convey meaning.



15000
BCE

Egyptian Murals

Large, pictographic murals communicate complex ideas to crowds of thousands. Hieroglyphic symbols—functioning as both representative images and phonetic components—augment larger images to blend visual and verbal communication.



3000
BCE

Public Speaking

The Greeks pioneer the study and practice of oratory and logography. Centuries later, Ars Oratoria (the art of public speaking) is a mark of professional competence in Rome, especially among politicians and lawyers.



500
BCE

Stained Glass Windows

Before the printing press, the Roman Catholic Church conveyed stories of saints and biblical characters to a mostly illiterate public through the colorful medium of stained glass. The messages stick.



950
CE

Bar Graphs

Bishop Nicole Oresme creates a "Proto-Bar Graph" for plotting variables in a coordinate system. Thankfully, he lacks distracting, modern textures.



1350
CE

Comic Strips

Swiss artist Rudolphe Töpffer develops the forerunner to today's modern comic strips; he tells complete stories using frames that contain both images and text.



1845
CE

Overhead Projector

Police begin using overhead projectors for their identification work, quickly followed by the military, educators, and businesses.



1945
CE

35mm Slide Presentations

The 35mm slide projector enables professionals to communicate ideas sequentially to larger audiences. The pioneering 35mm slide firm Genagraphics charges from \$300 to \$1500 per proprietary slide.



1950
CE

PowerPoint

The click heard 'round the world: PowerPoint 1.0 debuts for the Macintosh. Suddenly everyone can design slides. Little consideration is given to whether or not this is a good idea.



1987
CE

Pervasive PC

PCs sit on every desktop in the workplace and high-stakes business communications evolve from printed documents to digital presentations. The 35mm slide companies go extinct almost overnight.



1992
CE

Cognitive Style of PowerPoint

Edward Tufte authors "The Cognitive Style of PowerPoint." In it, he suggests that PowerPoint impaired the quality of the engineers' investigative analysis on the Columbia Space Shuttle when it was gravely impacted by debris.



2003
CE

An Inconvenient Truth

Al Gore raises environmental consciousness, wins an Academy Award, and receives the Nobel Peace Prize for telling a compelling story about climate change with little more than a slide show.



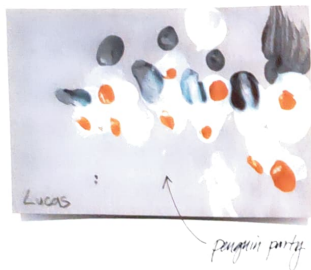
2007
CE

When you think of presentations, your immediate thoughts probably travel only as far back as 1987—the beginning of the PowerPoint era. If you broaden your perspective, you might recall an age of 35mm slides and flip charts—the latter half of the last century. And though the means and methods have changed over time, the messages by and large have not: you recount stories, present new information, strive to change others' minds. The world is wired for visual as well as verbal communication. Don't believe it? Consider this timeline:

Don't Commit Career Suiside

We are all inherently visual communicators. Consider kindergarten: crayons, finger paints, and clay propelled our expression, not word processors or spreadsheets.

Yet, the stories you told with these limited means were at least as good—perhaps better—than what you can accomplish with today's technology. For instance, the following image was painted by Lucas, the son of a friend of mine. On first glance it looks like meaningless globs of paint. But Lucas' story is about a penguin party, and in that context, the images make brilliant sense. The greater message here is that stories are how people understand and relate to the world, and they naturally associate those stories with appropriate imagery.



Unfortunately, somewhere, at some time, someone probably told you that you weren't very good at drawing. And, after looking around and comparing yourself to other kids in the classroom, you probably consented, threw in the towel, and decided that piano lessons or football might prove a better bet for primary education glory.

Now, as an adult, you may not try anymore—at least in the visual realm. This is ironic considering that your employers and colleagues assess you by how well you communicate—a skill that is reflected in annual reviews, pay increases, promotions, and even your popularity. Effective communication is a job requirement now, whether you're trying to beat competitors, communicate vision, demonstrate thought leadership, raise capital, or otherwise change the world. And like it or not, your profession likely requires you to communicate using a visual tool, regardless of your proficiency or training in this medium. Business schools in particular drill their students in management, accounting, and technology, but few offer anything approaching Design 101—the one thing that combines creative thinking, analytics, data assimilation, and the inherent ability to express oneself visually.

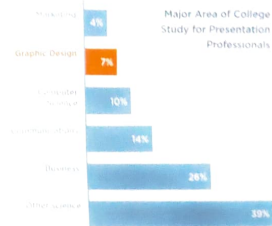
Others have noted the Catch-22 pressure of being able to communicate well visually without the proper training. Marcus Buckingham, on a conference call about his book *Go Put Your Strengths to Work*, recounted his own experience this way:

"I figured out pretty quickly that most of the sales people I was working with weren't very good at putting together PowerPoint presentations. I took it upon myself to become as expert as I could. Not that I'm some genius now, but I can put together a great presentation really quickly and effectively. And it made me a more valuable asset."

Marcus Buckingham
Author, *Go Put Your Strengths to Work*

Closer to home, my brother-in-law, a retired lieutenant commander in the U.S. Navy, recalled that presenting less than optimum slides made promotion difficult. Twenty years ago, no one would have guessed that knowledge of this visual medium would be so pervasive or so important. Indeed, International Communications Industries Association concluded from a recent study that very few presentation professionals themselves have had any graphic design training. And these are the people who work at large companies and build slides full time!

Making bad slides is easy, and it will negatively impact your career. Invest in your slides, but invest in your own visual skills as well. The alternative is to inadvertently commit career suiside.



Source: ICA, 2000 survey

A Case for Presentations

Corporations spend hundreds of millions of dollars on advertising, marketing, and PR to attract and retain consumers. They run TV ads, place banner ads, erect billboards, write articles, and dispatch massive amounts of annoying junk mail, all to persuade customers of the superiority of their products and services. Sometimes this is enough: consumers are convinced and accounts are won. But other times, these expensive means are merely a prelude to a personal engagement—one that will depend on the effectiveness of a presentation to seal the deal.

Now, consider the disparity between the content, design, and production values of that \$100 million campaign, and the slide show residing on your laptop. Consider also that this slide show may be the last engagement you have with your customers before they make a purchase decision. From an experiential standpoint, few things could be more anticlimactic than a massive campaign followed by an unorganized, unimpressive presentation that might not be relevant to what the audience needs from you or the company.

How is it that companies became so focused on a grandiose approach to marketing yet so reluctant to spend even a fraction of the time needed to create a great presentation?

Truth be told, the reason many organizations relegate slides to the bottom of the marketing food chain has to do with how they approach brand.

Many companies have forgotten—or simply never realized—what branding is. Rather than a name or logo or tagline that reflects what a company thinks of itself, brand is what a company stands for in the hearts and minds of its customers; to be successful, the company must have an emotional connection with the consumer.

Similarly, presentations all too often reflect the agenda of the presenter rather than build a connection with the audience. This is unfortunate because presentations could be considered the last branding frontier, in terms of both the attention paid to them and where they fit in the sales cycle.

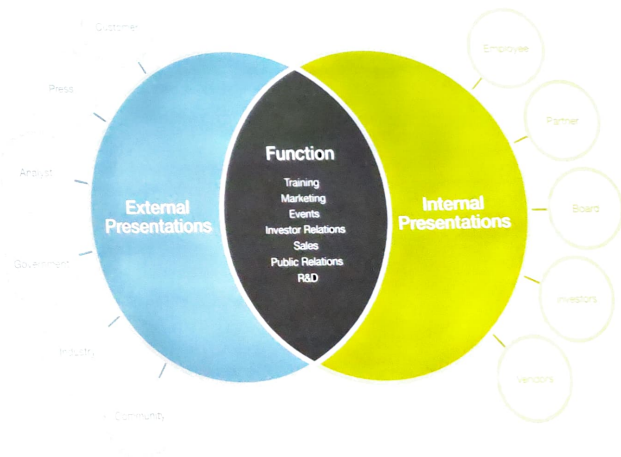
In many instances, presentations are the last impression a customer has of a company before closing a business deal.

Indeed, it wouldn't take much for any company to stand out from its competitors if it paid some respect to its brand—and its audience—through its presentations.

The collective (mis)use of this presentation medium has informed everyone's opinion of it. Some people simply don't understand how powerful and moving a presentation can be.

Let's go fix that.

Presentations are a tool for high-stakes internal and external communications. This medium will influence many of your important constituents and the impression they develop of you and your company.



So Where Do You Begin?

Use the right tool the right way.

At a certain point, the number of words on a slide prevents it from being a visual aid. You've been there. The audience is reading the slides instead of paying attention to the presenter, the presenter is reading the slides instead of connecting with the audience, and the whole endeavor would have been better served through a well-composed document or even an email.

Unfortunately, the negative habits that lead to this kind of presentation style are deeply rooted and difficult to change. But consider this: audiences will either listen to what a presenter is saying or read the slides themselves. They won't do both. Why? People tend to focus on one stream of verbal communication at a time—listening and reading are conflicting activities.

On the other hand, it's natural for people to simultaneously pay attention to both verbal and visual communication. That's why great slides serve as a visual aid that reinforces the speaker's message.

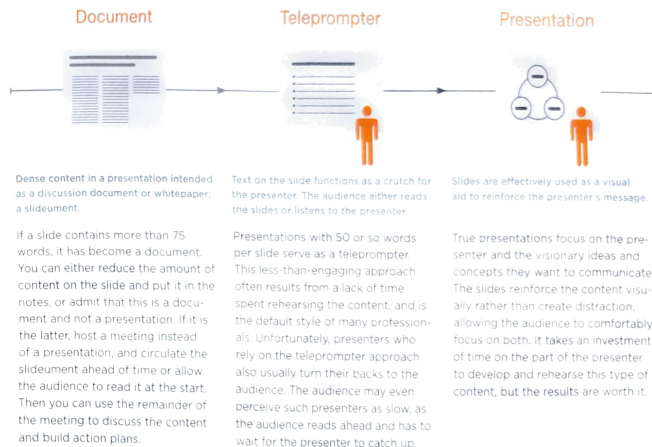
However, the heavy use of text occasionally might be appropriate, but in such a case, you should be careful what medium you use. Too often, presentation software is used to create documents. Garr Reynolds, author of *Presentation Zen*, calls these slideuments. So before your next presentation, assess how you've used the application. Did you create a document or a presentation? Either adjust your documents into slides or trust that your audience is smart enough to read and circulate your slides as a document.

"Communication is about getting others to adopt your point of view, to help them understand why you're excited (or sad, or optimistic, or whatever else you are). If all you want to do is create a file of facts and figures, then cancel the meeting and send in a report."

Seth Godin,
Author, *Really Bad PowerPoint*

People will love you for respecting their time enough to use the media appropriately.

This book primarily focuses on presentations that are on the far right of this spectrum.



The audience will either read your slides or listen to you. They will not do both. So, ask yourself this: is it more important that they listen, or more effective if they read?

The Presentation Ecosystem

Jim Endicott, author of *The Presentation Survival Skills Guide*, refers to the presentation development process as a three-legged stool—message, visual story, and delivery.

As a presenter, you rely on the interdependence of your ideas, graphics, and execution. As an audience member, how many times have you attended a presentation only to wonder, after it's too late to extricate yourself gracefully, why you're there? The presenter may be very well-prepared and as fired up as a televangelist, but it's impossible to figure out the intent and why it's worth caring about. Most likely, the presenter focused on their "message" leg—dumping everything they know onto individual slides—but forgot to give equal attention to developing the "visual story" and "delivery" legs of the stool.

Presenters often read their slides instead of putting in the effort necessary to transform them into visual stories that support their message.

Today's presentation graphics tend to distract audiences. The result might be described as visual vertigo: audiences are jarred by having to resolve graphical disparities and dissonances that arise from poorly rendered perspectives, inconsistent lighting sources, and exasperating animations. Whether or not the content and delivery are good, people exposed to crudely constructed media will walk away from a presentation subtly agitated and thus less receptive to the message. Even worse, visuals devoid of clarity can cause a subliminal lack of trust. The presenter doesn't realize that the audience members care solely about what the presenter can do for them.

You also need to better anticipate the audience's needs initially and adjust the delivery accordingly. Presentations are quickly moving from face-to-face to exponentially more powerful mass media. How many times have you developed a presentation with the sole intent of delivering it in-person—only to find out later that a much larger audience will view it online? You'll need to modify that same content so that it resonates with its web-based audience.

This is a sensitive ecosystem. Striking a harmonious balance is important. Odds are high that you've been on both sides of the podium. As presenters, the natural tendency is to stay in your own zone, thinking more about how successful you'll feel instead of how your content, visuals, and delivery will be perceived.



Responding to Audience Needs

Before beginning a presentation, it's important to ask yourself questions about your audience. Who are they? What are their needs, and how can you address them? How can the information you have make their lives better? What do you want them to do after the presentation is over? Questions like these are critical to developing relevant, resonant content.

Thinking about the audience will ensure you are placing their needs first, and give you a benchmark against which you can check your message.



Describe your audience, including some combination of their fears and needs, and state the opportunities and challenges you need to address in your communications. And of course, don't overlook their state of mind when they sit down to hear your message.

Most importantly, share the benefits of your position and provide the audience something to do after the presentation—a call to action. Whether your presentation goal is to share information or recruit people to a cause, you are faced with the same challenge of getting the audience to follow through. What should they do with the information you've provided them? You can relay your message time and time again, but the audience can't act on the message until you tell them what's in it for them.

Those with the most to lose—and the most to gain—are the listeners. Define their needs, surpass their expectations, and turn them into agents of your cause. Without your audience, you are nothing.

Consider the kind of relationship you want to have with your audience. Do you want to be their hero? Their mentor? Their cheerleader? Like these characters, good presenters aren't in it for themselves; they're in it for others. Take note.

Consider this list of questions when trying to understand an audience. Remember, presentations and audiences may vary, but one important fact remains constant: the audience didn't come to see you. They came to see what you can do for them.

Seven Questions to Knowing Your Audience

- | | |
|------------------------------------|--|
| 1 What are they like? | Demographics and psychographics are a great start, but connecting with your audience means understanding them on a personal level. Take a walk in their shoes and describe what their life looks like. |
| 2 Why are they here? | What do they think they're going to get out of this presentation? Why did they come to hear you? Are they willing participants or mandatory attendees? This is also a bit of a situation analysis. |
| 3 What keeps them up at night? | Everyone has a fear, a pain point, a thorn in the side. Let your audience know you empathize—and offer a solution. |
| 4 How can you solve their problem? | What's in it for the audience? How are you going to make their lives better? |
| 5 What do you want them to do? | Answer the question "so what?" Make sure there's clear action for your audience to take. |
| 6 How might they resist? | What will keep them from adopting your message and carrying out your call to action? |
| 7 How can you best reach them? | People vary in how they prefer to receive information. This can include everything from the setup of the room to the availability of materials after the presentation. Give the audience what they want, how they want it. |



To ensure you keep your audience needs top-of-mind, download an audience persona slide.

Finding Your Inspiration

Much of communication today has the quality of intangibility. Services, software, causes, thought leadership, change management, company vision—they're often more conceptual than concrete, more ephemeral than firm. And there's nothing wrong with that. But it's a struggle communicating these types of ideas because essentially they are invisible. It's difficult to share one's vision when there's nothing to see.

Expressing these invisible ideas visually, so that they feel tangible and can be acted upon, is a bit of an art form. The best place to start is not with the computer. A pencil and a sheet of paper will do nicely.



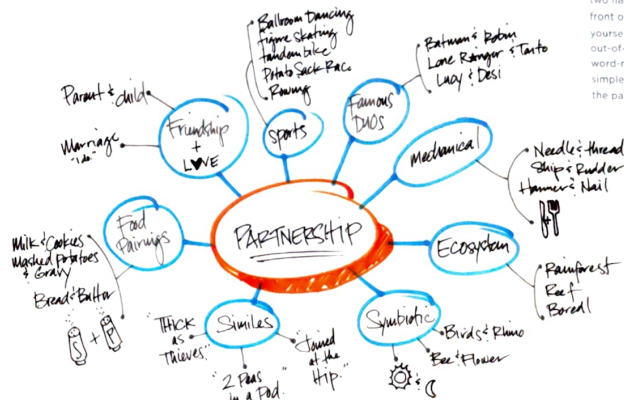
Change your environment when you need to be creative. Find a spot away from your desk. Yeah, Salvador Dali was a bit eccentric, but try writing your next speech in the tub with paper and pencil and just see if it's not better.

Why take this seemingly Luddite approach? Because presentation software was never intended to be a brainstorming or drawing tool.

The applications are simply containers for ideas and assets, not the means to generate them.

It's been an easy trap to fall into, launching presentation applications to prepare content. In reality, the best creative process requires stepping away from technology and relying on the same tools of expression that you grew up with—pens, pencils, and crayons. The goal is to generate ideas—not necessarily pictures yet—but lots of ideas. These can be words, diagrams, or scenes; they can be literal or metaphorical; the only requirement is that they express your underlying thoughts. The best thing about this process is that you don't need to figure out how to use drawing tools or where to save the file. Everything you need you already have (and don't say you can't draw; you're just out of practice). This means you can generate a large quantity of ideas in a relatively short amount of time. And that's your primary objective when generating ideas: to create a large quantity of them.

And you wondered where all Dali's liquid inspiration came from



Cheesy metaphors are a cop-out. Instead of going for the photo of two hands shaking in front of a globe, push yourself to generate out-of-the-box ideas by word-mapping. Here's a simple example around the partnership theme.

Often ideas come immediately. That's good, but avoid the potential pitfall of going with the first thing that comes to mind.

Continue to sketch and force yourself to think through several more ideas. It takes discipline and tenacity—especially when it feels like you solved it on the first try. Explore words and word associations to generate several ideas. Use word-mapping techniques, where you write or draw word associations like in the example above. Digital natives might prefer mind-mapping software for this phase. Stronger solutions frequently appear after four or five ideas have percolated to the top. Continue generating ideas even if they seem to wander down unrelated paths. You never know what you might find, after all.

Once you've generated an enormous amount of ideas, identify a handful that meet the objective of the vision or concept you're trying to communicate. It matters less what form they take at this point than that they get your message across.

Take the time and spend the creative energy because the payoff will be a presentation that people not only remember, but one that inspires them to action.

Innovating with Sticky Notes

Great inventions combine new technology with widespread applicability. In the case of the Post-it, a scientist at 3M applied low tack adhesive to paper and used it to mark his hymnal for choir practice at church. That's how this favorite brainstorm medium was born.

Sticky notes allow ideas to be captured, sorted, and re-arranged as needed.

When generating ideas, one idea per sticky note is preferable. And use a Sharpie. The reason? If it takes more space than a Post-it and requires more detail than a Sharpie can provide, the idea is too complex.

Simplicity is the essence of clear communication. Additionally, sticky notes make it easy to re-order content until the structure and flow feel right. On the other hand, many people prefer a more traditional storyboarding approach, preferring to linearly articulate detailed ideas. That's fine, too. The point is not to prescribe exactly how to work, but to encourage you to generate a lot of ideas and to do so quickly.

For other ideas in content development methodologies, check out Decker Communication's grid at www.decker.com and Cliff Atkinson's *Beyond Bullet Points* at www.beyondbullets.com.



TIP

When brainstorming, don't settle for the first solution. Think through multiple alternatives. Don't be stingy with the sticky notes.

Whatever your brainstorming method, remember the following guidelines.

Rule #1:

POSTPONE
& WITHHOLD
your judgment
of **IDEAS**

Rule #2:

ENCOURAGE
WILD
EXAGGERATED
ideas

Rule #3:

Quantity
Counts
at this Stage,
NOT Quality

Rule #4:

**BUILD ON
IDEAS**
put forward
BY OTHERS

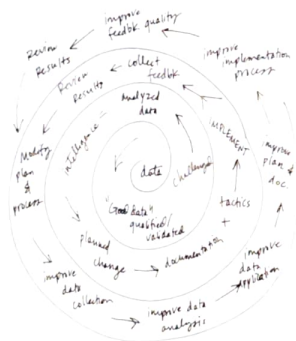
Rule #5:

**every
person's
every IDEA
has = worth**

After you've created an enormous amount of ideas, classify them, challenge them, and filter them down to a core set of meaningful ideas that will resonate.

Collaborating to Get Clarity

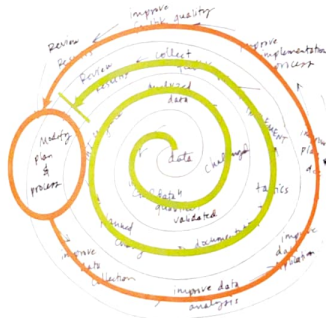
Often, ideas that seem clear and straightforward in one's own head can be confounding to others. That's why it's important to explore multiple ways of visualizing an idea. Don't rely solely on words to get your point across either; make sketches of what you see when you are speaking.



WHAT HE SAID

In visualizing his ideas, one executive sketched this diagram. It reads as a spiral or a single, linear flow.

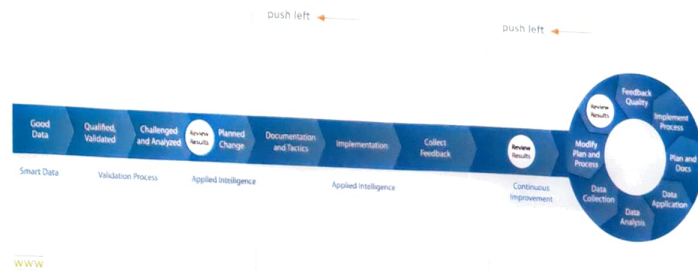
Then, bring in a colleague or two and walk them through the sketches. Ask them to listen not only to your words but to evaluate whether or not the sketches capture your intent. If they become confused while you are talking, have them ask clarifying questions or sketch out their own version of your ideas.



WHAT HE MEANT

After considering the executive's intent, two types of flow emerged: the process begins as a linear sequence from the center, but at the point he got to "modify, plan, and process" it turns into a circular flow.

- Occurs once in process
- Continues in a cycle



www

WHAT THE DESIGNER CREATED

The flow splits over a series of three slides. Using a push transition, it reveals the multifaceted process. The content from the green line on the left is a linear flow occurring in the first two slides above. The third slide relates to the orange line and is expressed as a circular flow. Splitting the content across three slides made it legible and ensured that each phase of the process received the proper focus.

TIP

It's important to test drive your presentation with colleagues before taking it out on the road.

Classifying Diagrams

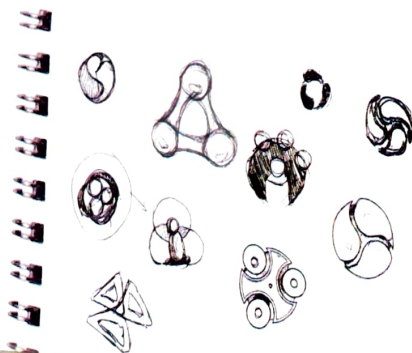
At my firm, we use sketchbooks to generate many ideas around a concept. I collected the sketchbooks of all the employees and created iconic representations of the most common diagrams. I cut these thousands of diagrams into little cards and sat on the floor for hours and sorted them. I felt like a little kid again.

After a long day hunched over the diagrams, a pattern and classification began to emerge. There were distinct differences in the diagrams and how they would be used to show relationships in information for professional communication.

A diagram is a good way to explain how parts of a whole interact. It's nearly impossible to communicate today without using shapes to symbolize various types of relationships and their interactions.



When I was little, I would sit on the floor for hours sorting and arranging objects, textures, and layers in a way that told a story.



Relationships, patterns, and classifications of the diagrams emerged from the sketches.

The diagram section depicts examples of six common diagram types. The first four types show common shapes that can be used to explain various abstract relationships. The last two types show illustrated solutions of a more literal, realistic nature.

Below is a key for how the section is organized.

Abstract Concepts



Flow

Linear
Circular
Divergent/Convergent
Multidirectional



Structure

Matrices
Trees
Layers



Cluster

Overlapping
Closure
Enclosed
Linked



Radiate

From a point
With a core
Without a core

Realistic Concepts



Pictorial

Direction
Location
Reveal
Process
Influence

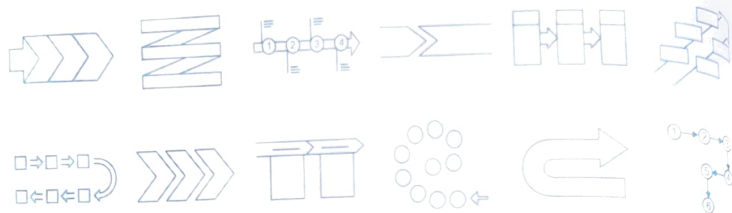


Display Data

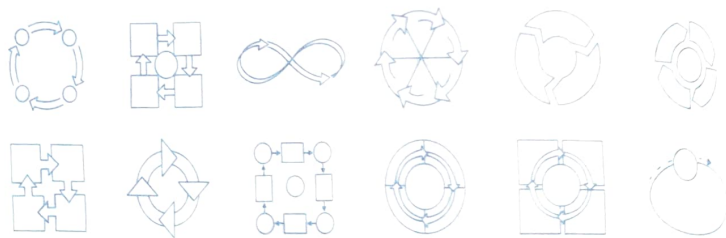
Comparison
Trend
Distribution

Abstract Concepts: Flow

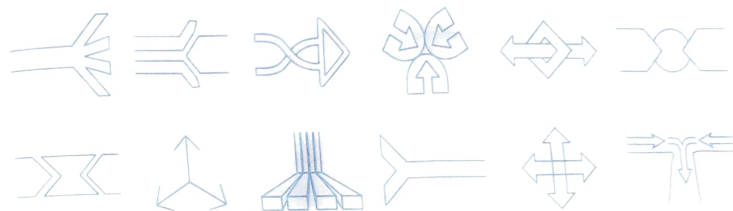
Linear: Flow construction that illustrates a process with a definite start and end point. The diagram can follow a straight line or be a series of steps along a line.



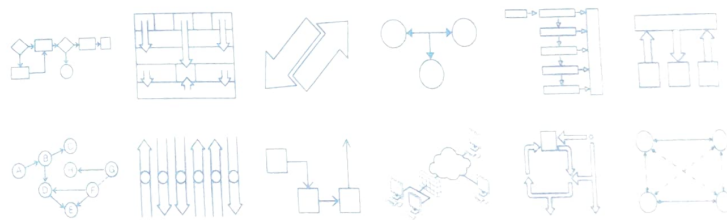
Circular: Flow that represents a continuous process without an end point. Any closed loop shape could work.



Divergent and Convergent: Flow that occurs when two or more elements either collide or separate out from each other as if splitting off.

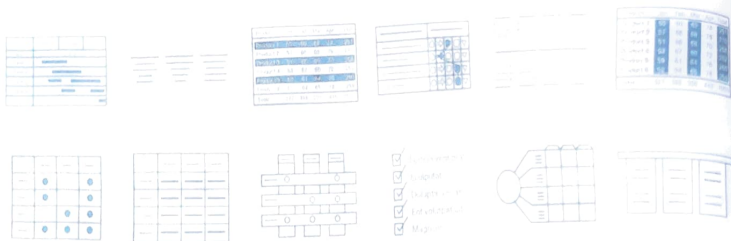


Multidirectional: Flow that expresses complex relationships—flow charts, for instance. Often these flows result from a combination of the preceding types.

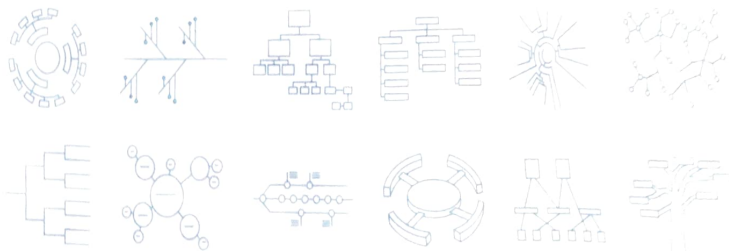


Abstract Concepts: Structure

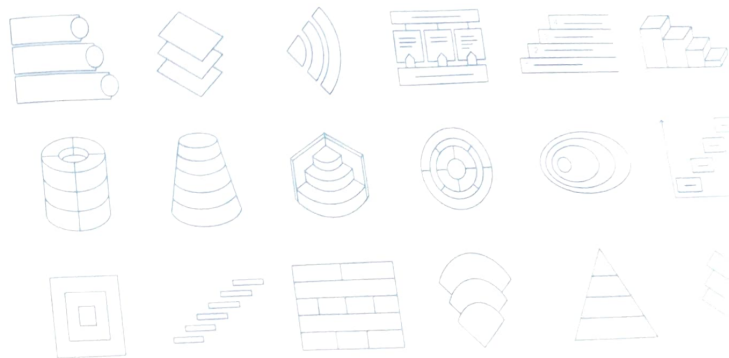
Matrices: Structures that compare data with at least two different data sets. One set of data can even be a yes/no set of data, as in a checklist.



Trees: Structures that indicate clear hierarchy. Relationships can be expressed between any number of objects.



Layers: Structures that show elements that stack or build on each other. They can depict both hierarchy and sequence.



Abstract Concepts: Structure

Matrices: Structures that compare data with at least two different data sets. One set of data can even be a yes/no set of data, as in a checklist.

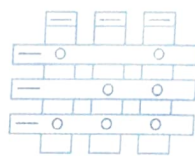
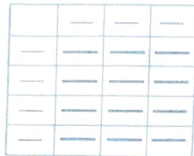
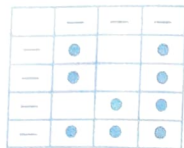
	Feb	Mar	July
Task 1	Task 1	Task 1	Task 1
Task 2	Task 2	Task 2	Task 2
Task 3	Task 3	Task 3	Task 3
Task 4	Task 4	Task 4	Task 4
Task 5	Task 5	Task 5	Task 5



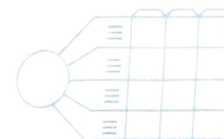
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Product 1	55	69	62	74	259
Product 2	57	68	68	79	272
Product 3	61	65	69	70	265
Product 4	53	67	60	72	252
Product 5	65	61	64	76	266
Product 6	52	64	67	75	258
Total	322	430	437	475	1664



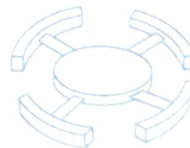
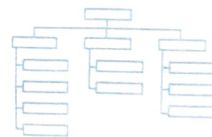
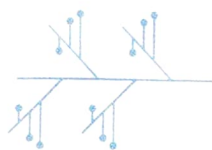
Product	Jan	Feb	Mar	Apr
Product 1	55	69	62	74
Product 2	57	68	68	79
Product 3	61	65	69	70
Product 4	53	67	60	72
Product 5	65	61	64	76
Product 6	52	64	67	75
Total	322	430	437	475



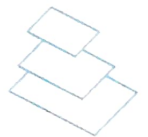
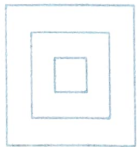
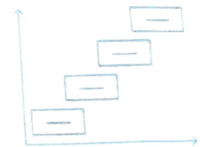
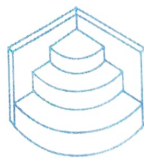
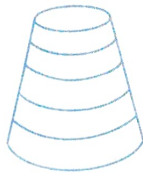
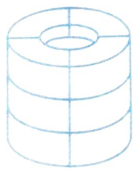
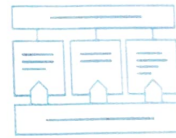
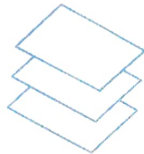
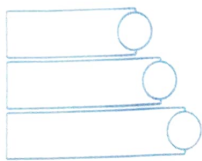
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Trees: Structures that indicate clear hierarchy. Relationships can be expressed between any number of objects.

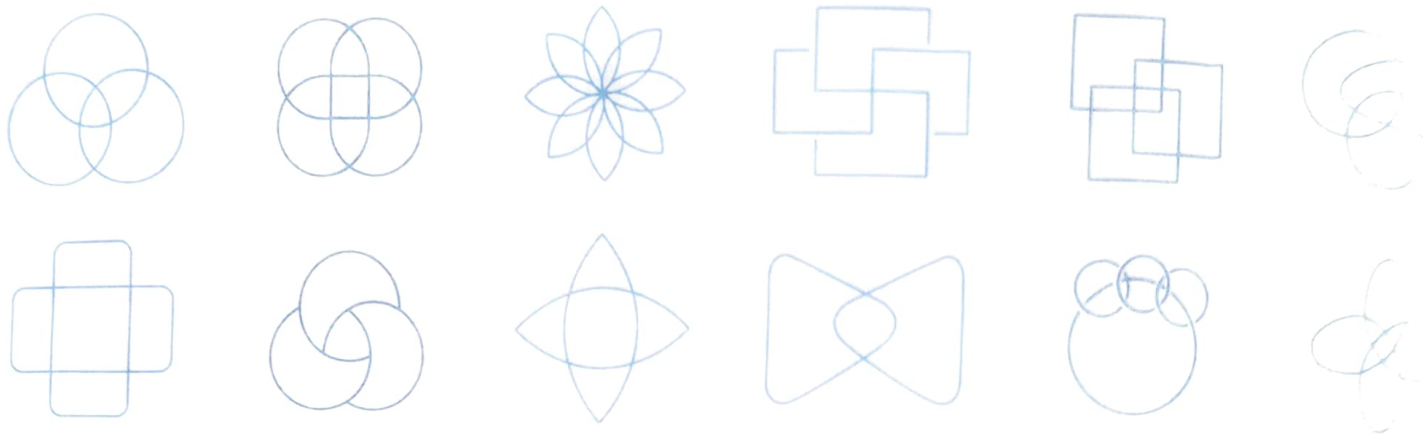


Layers: Structures that show elements that stack or build on each other. They can depict both **hierarchy** and **sequence**.

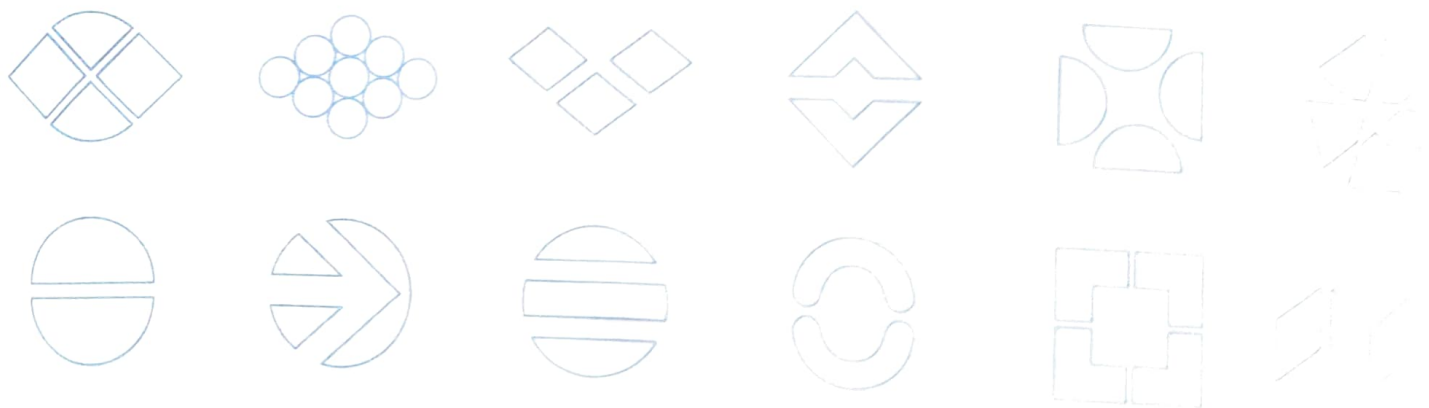


Abstract Concepts: Cluster

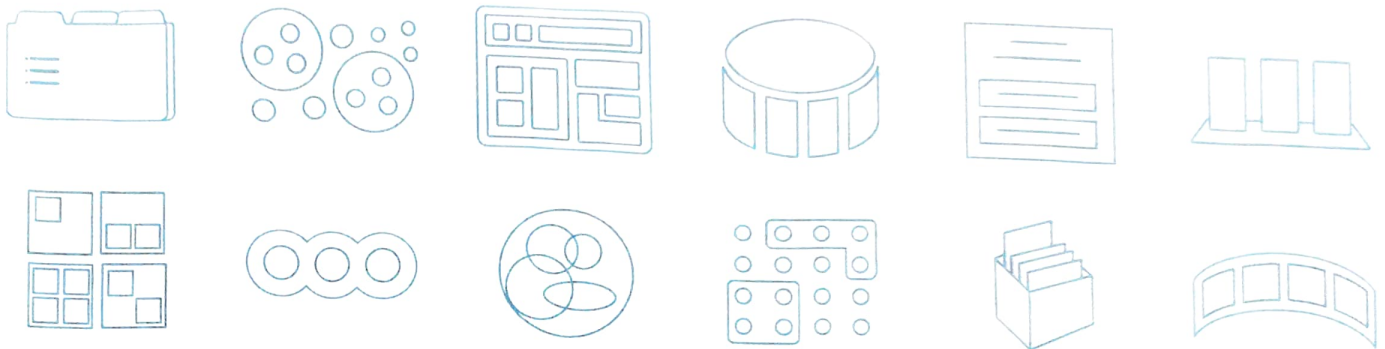
Overlapping: Clusters that overlap and indicate shared sets, interest, or responsibility. Sometimes they form a new shape or area within the overlap.



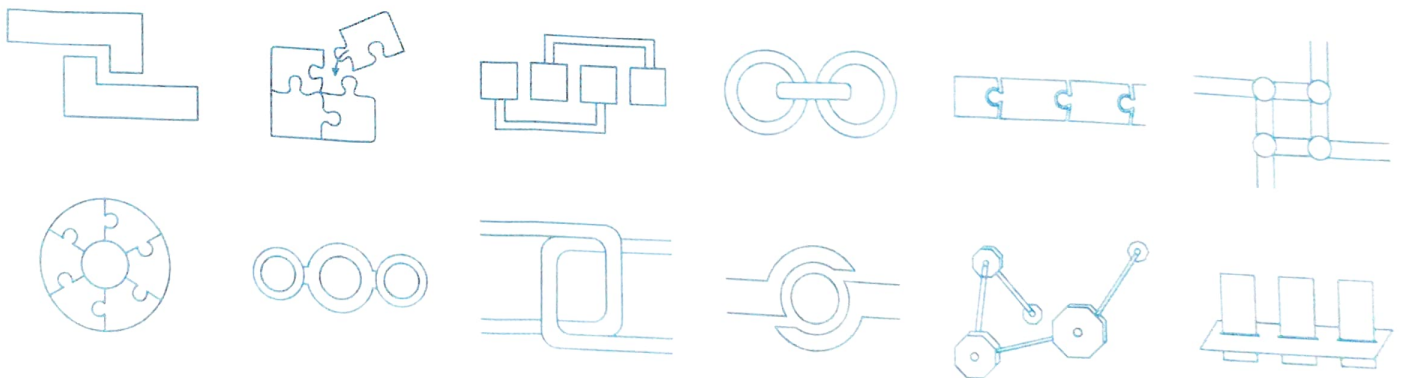
Closure: Clusters that emerge when shapes combine to create another shape. This principle of Gestalt psychology can be useful when expressing the idea that "the sum is greater than the parts."



Enclosed: Clusters that are enclosed and contain at least one element that envelops another. Grouping in this way indicates which elements are part of a higher order, and which stand alone.

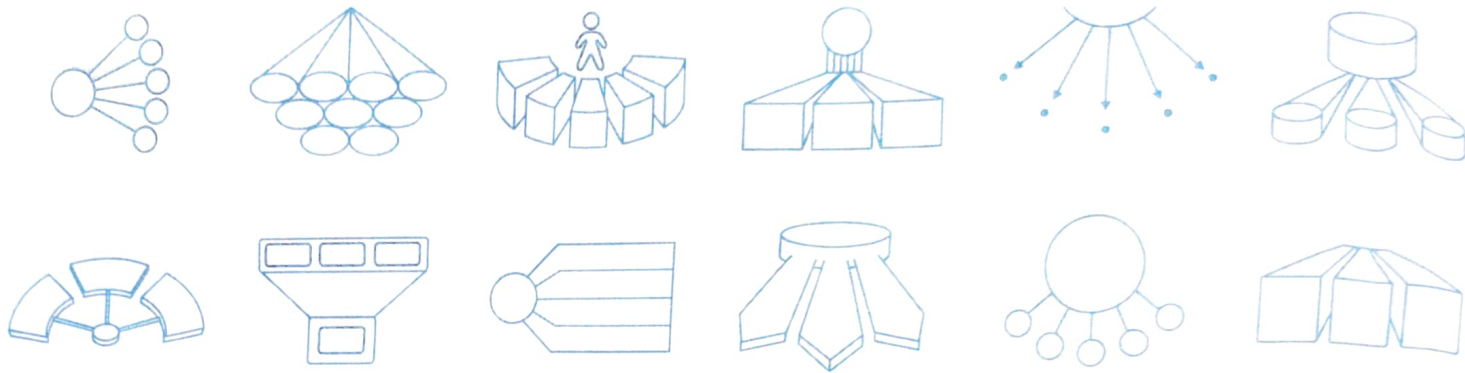


Linked: Clusters that are linked as a unifying element to group items. A unifying element links related groups of items. It could be a line, shape, or connector of any kind.

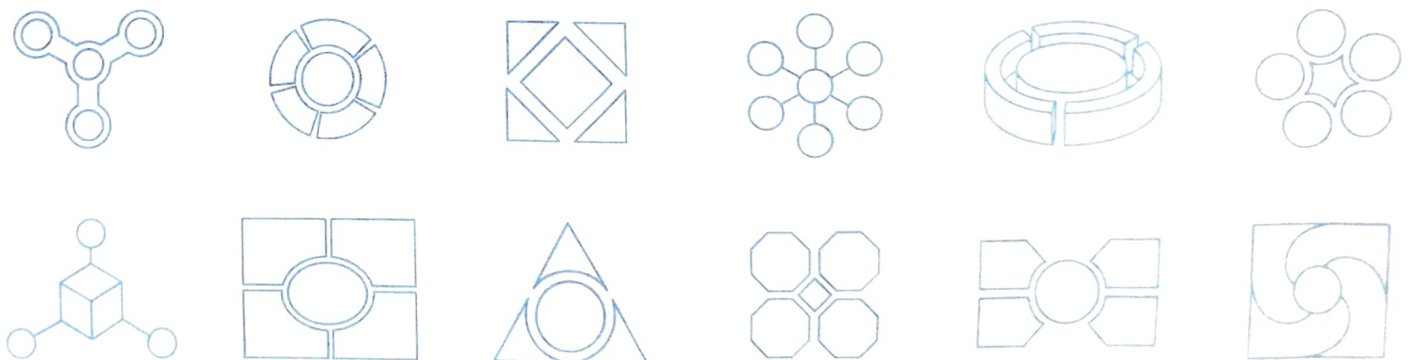


Abstract Concepts: Radiate

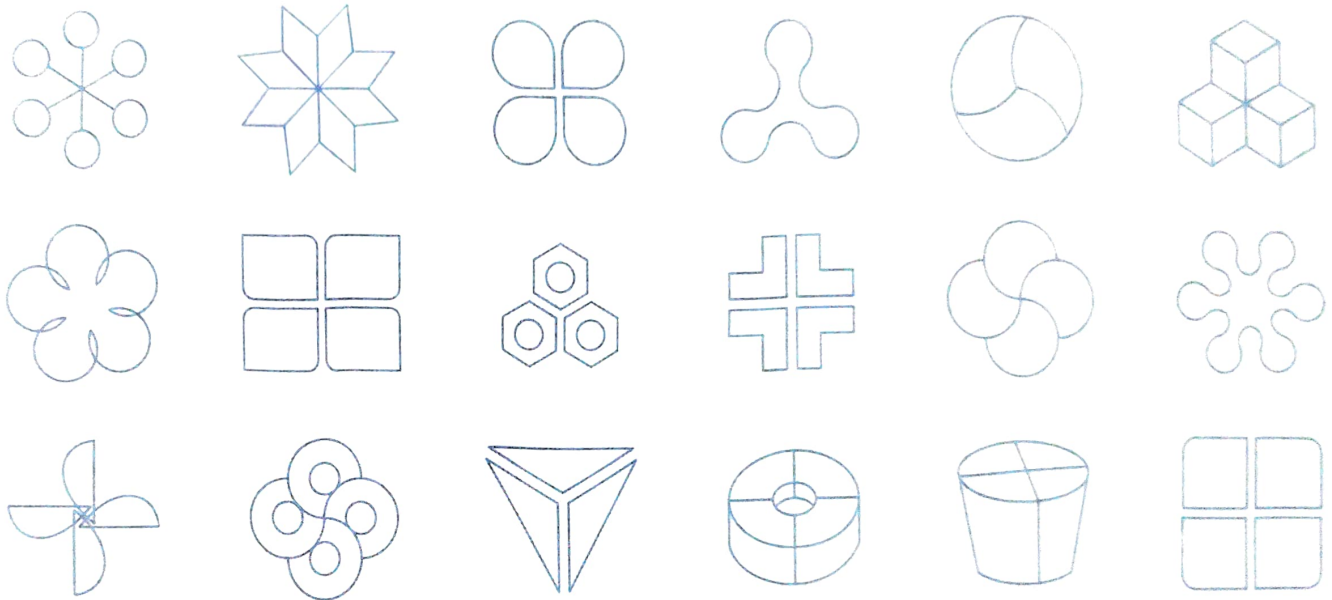
From a point: Occurs when a single directional "burst" emerges from either a graphic or point that has a clear point of origin.



With a core: Creates a parent-child relationship. The outer elements connect with a central element to hold the family together.



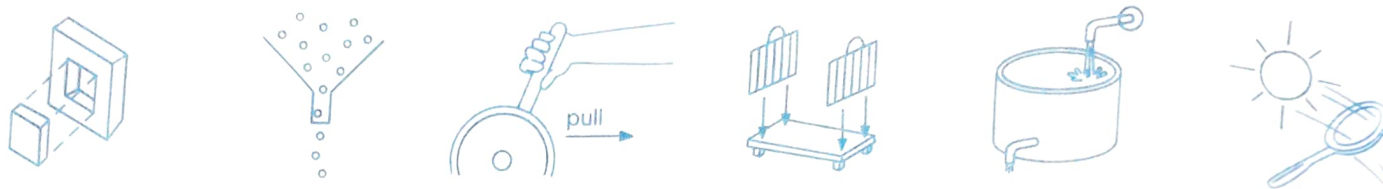
Without a core: Implies that elements connect through proximity or mutual attraction. They are tied to one central area.



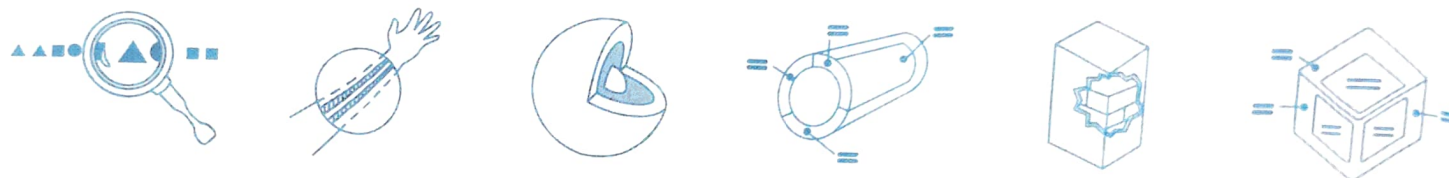
You've noticed that the abstract concepts are usually shapes that are combined to show relationships. In this next section you'll see a sampling of diagrams that are realistic. It is by no means exhaustive. There's no limit to how you tell your story visually.

Realistic Concepts: Pictorial

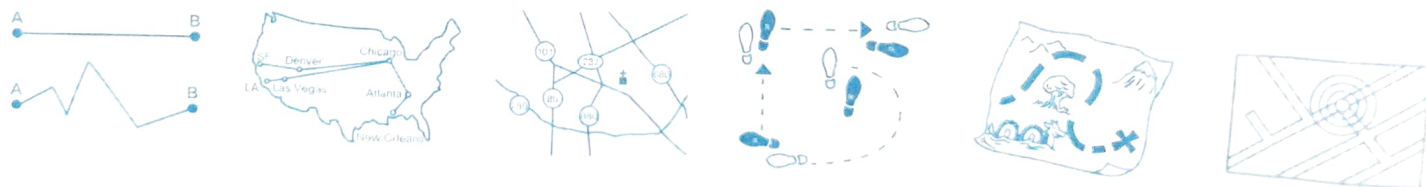
Process: A snapshot of how things work sequentially as a product or system.



Reveal: An illustration of hidden information shown by slicing, peeling, or otherwise exposing a thing's inner workings.



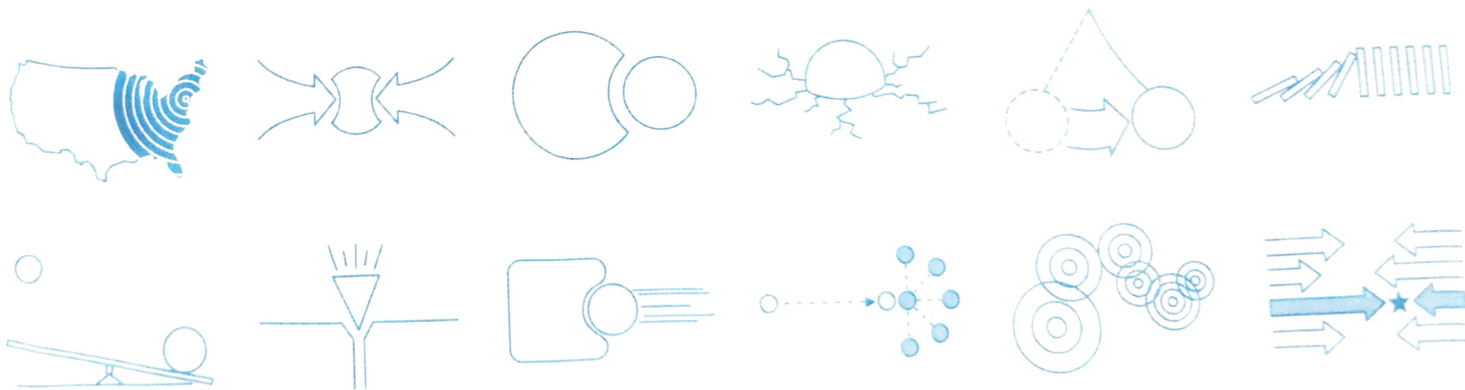
Direction: Showing where to go or how to get somewhere. They usually feature a starting location and destination, and may contain written directions.



Location: Showing where something is in the context of geography, topography, system, or shape. These diagrams call attention to a specific location while providing context.

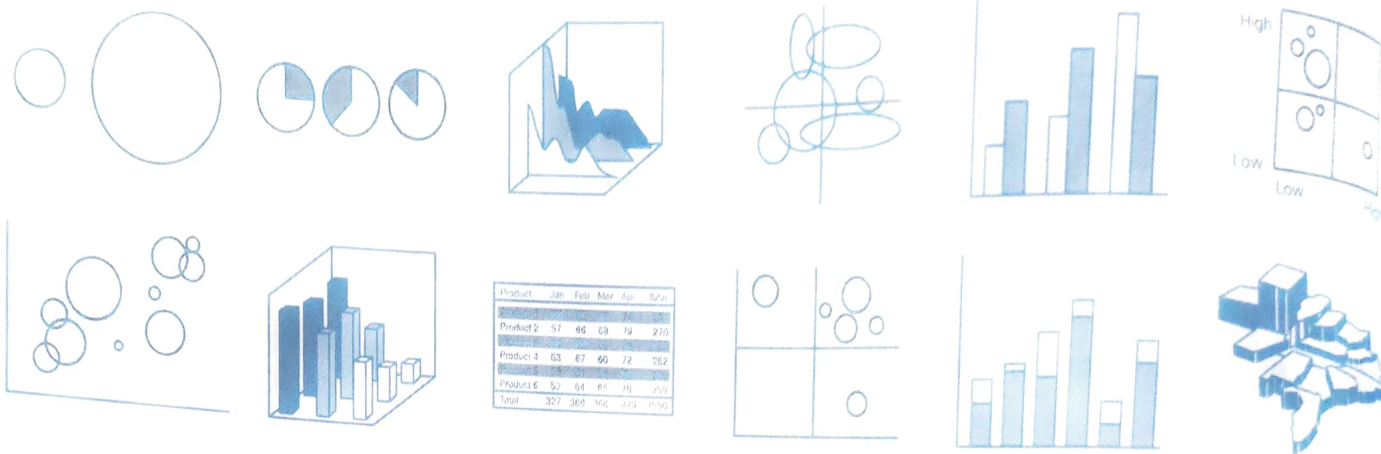


Influence: Demonstrating the resulting impact of various interacting elements.

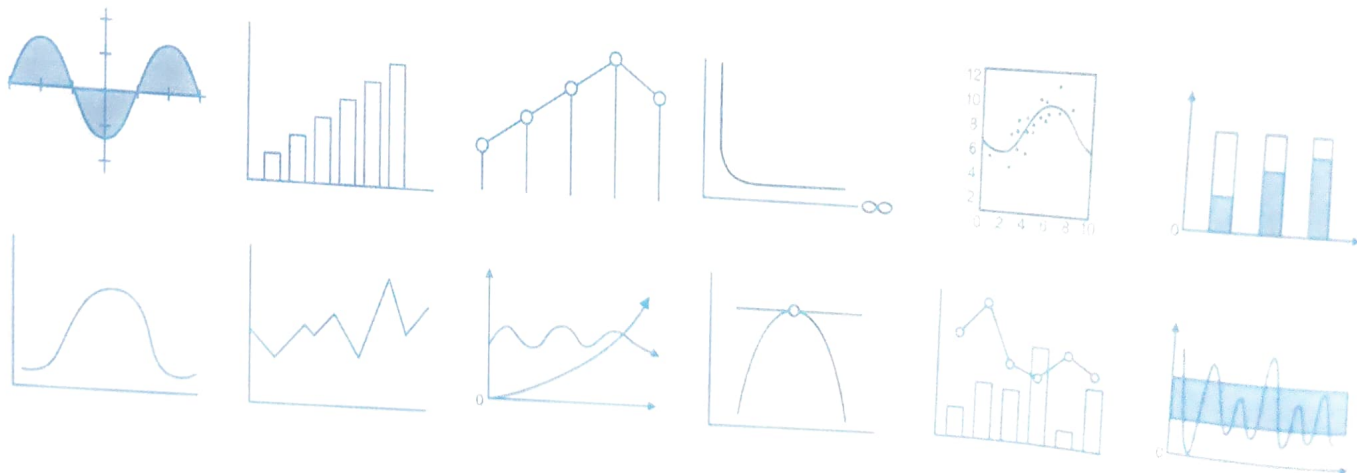


Realistic Concepts: Display Data

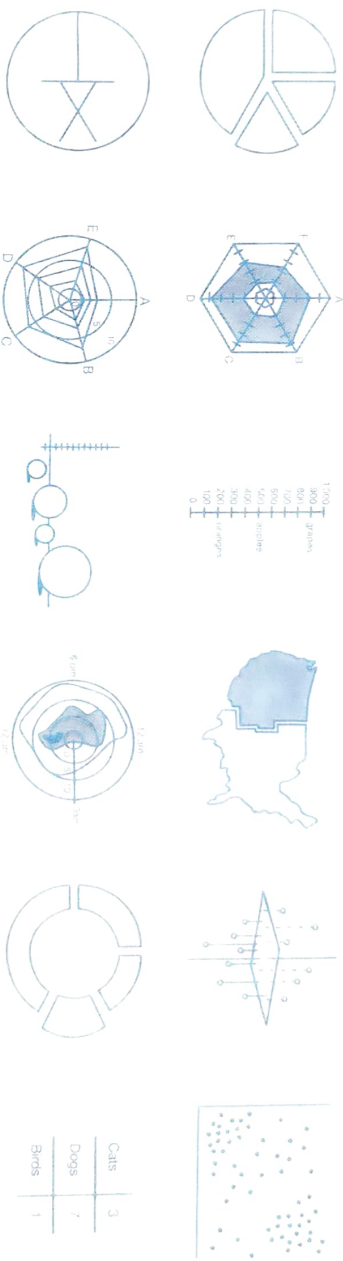
Comparison: Juxtaposition of two or more sets of information to illustrate differences. Bar graphs, pie charts, and any number of other methods are suitable.



Trend: One parameter of the data represents time, to indicate a trend. Change over time is the most important aspect of these diagrams.



Distribution: Expression of a pattern in seemingly disparate data. Whether a scatter plot, bell curve, or other model, distribution diagrams correlate singular instances into a larger pattern.

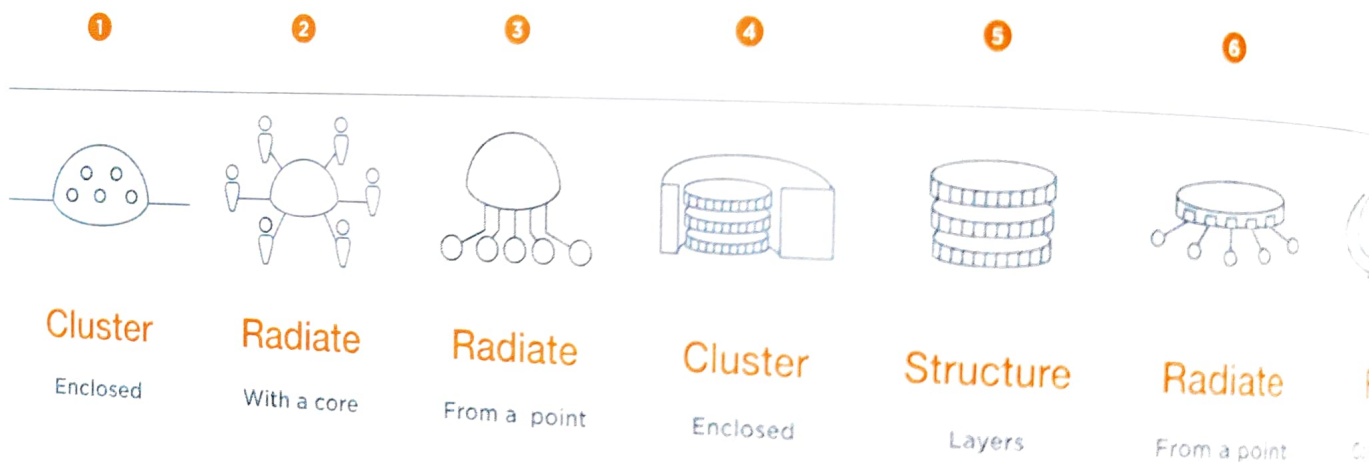


Including display data in the diagram section may seem illogical. But data displayed visually is a diagram of sorts—and it's often more clear when the display highlights the meaning of the information. Rather than oversimplify the complexities of statistics, successful information design can often incorporate multiple parameters, telling a richer story of cause-and-effect than data points alone.

Making Diagrams Work Together

Now that you've seen the building blocks for abstract and realistic concepts, you'll look at how to build them into a complex concept. Complex concepts can be communicated by connecting more than one diagram type. In the Silicon Valley, there are many presenters who need to explain software structures, web applications, and other invisible concepts visually. On the facing page is a concept sketch of a data warehouse solution that the customer called a "marketecture" diagram. It needed to show how the products and process work together like a "system."

A system similar to the one shown is a good way to represent elaborate and complex relationships. System diagrams can also show how things work or where things are. When presenting these complex relationships, remember that you want your audience to process one idea at a time—so this type of diagram needs to build over time.



Following the Five Data Slide Rules

When it comes to displaying data in your presentation, you must adhere to one principle above all others: clarity.

Projecting your data on a slide puts you at an immediate disadvantage to printing it in a white paper or scientific text. In a presentation, the audience doesn't have the benefit of being able to pull your data in close to examine it.

So it is absolutely crucial that any data in your presentation carries with it a clear message. And you can keep that message clear by remembering this one fact:

**Data slides are not really about the data.
They are about the meaning of the data.**

Most presenters don't really understand this distinction. How many times have you sat through presentations where the speaker referred to one complicated chart for five minutes, after which you still couldn't figure out the point of the slide?

The problem is that, most of the time, slides aren't a good medium for showing complex data. When it's important for your audience to examine the data and come to their own conclusions, you should distribute the data as a printout.

A cluttered or overwhelming data slide can derail even the most compelling speaker, so only show data in your presentation if the data helps you better illustrate your conclusions. And be sure to display it in a way that the audience can absorb easily (even from the cheap seats).

Use the following five principles to present your data in the clearest possible way:

- 1 Tell the truth
- 2 Get to the point
- 3 Pick the right tool for the job
- 4 Highlight what's important
- 5 Keep it simple

Telling the Truth

As a presenter, nothing commands like credibility. This is especially true when it comes to presenting data. Though it's unlikely that all the data can be clearly depicted on a single slide, successful presenters treat the data they do show with absolute integrity.

Additionally, be prepared to provide access to the complete data set, if requested, and be ready to answer questions about your conclusions.

Your audience deserves consideration here, too. Audiences that are analytical, scientific, or engineer-minded tend to look at data with a skeptical eye—it's what they're trained to do, after all. If your data has been boiled down or clarified too much, they may feel it has been manipulated, has become "marketing data," or is no longer substantial. To prevent these assumptions, avoid decorating your data; ornamentation can detract from credibility.

Getting to the Point

What is it you want your audience to “get” from your data? What’s the message you want them to take away?

To communicate your data effectively, you first must articulate the conclusions you want your audience to adopt.

Begin by asking yourself, “What would I like them to remember about this data?” Usually, this will be a point about causality, such as the conditions that led to the results in your data, or the future conditions that the data most likely will create. Whatever the cause, you need to draw some kind of meaning from the data so that you can effectively express that meaning to your audience.

Consider the example below. At first glance one might conclude that the main point of the slide is that revenue has increased. Granted, that is a major point, but not the primary point that the presenter wanted the audience to walk away with. In reality, this presentation was delivered by the Human Resources department to petition an executive leadership team to give their department a larger budget. Just as sales had hit a low point in March, the HR department had rolled out a new sales training program. No other organizational changes had happened in that timeframe and they wanted to correlate the spike with their program. The most important information therefore was the point in time where their program was implemented.



First impression on this chart is the enormous sales growth and recovery.



In reality, the purpose of this chart was for HR to draw attention to the March timeframe.

Once you are certain about the meaning of your data, you'll find it much easier to build the slide. Your conclusions will determine the type of graph you choose, and how (or if) you use emphasis.

These chart samples show that different types of conclusions can be drawn from the same data. Choosing what you want the audience to focus on is important so they process the information quickly.

Revenue Trends

	Q1	Q2	Q3	Q4	Total
Americas	-18%	7%	25%	2%	2%
Australia	47%	-7%	26%	15%	17%
China	16%	-5%	1%	7%	19%
Europe	57%	10%	-3%	7%	13%
India	57%	6%	-3%	7%	13%

This chart highlights Q1 to emphasize that this was the most successful quarter.

Revenue Trends

	Q1	Q2	Q3	Q4	Total
Americas	-18%	7%	25%	2%	2%
Australia	47%	-7%	26%	15%	17%
China	16%	-5%	1%	7%	19%
Europe	57%	10%	-3%	7%	13%
India	57%	6%	-3%	7%	13%

This chart emphasize that the Americas had the lowest revenue growth.

Revenue Trends

	Q1	Q2	Q3	Q4	Total
Americas	-18%	7%	25%	2%	2%
Australia	47%	-7%	26%	15%	17%
China	16%	-5%	1%	7%	19%
Europe	57%	10%	-3%	7%	13%
India	57%	6%	-3%	7%	13%

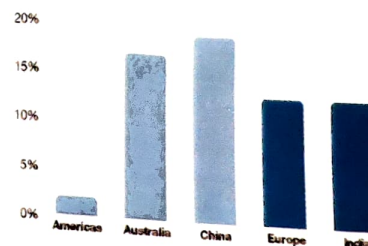
Here the focus is on when and where the revenue declined.

Revenue Trends

	Q1	Q2	Q3	Q4	Total
Americas	-18%	7%	25%	2%	2%
Australia	47%	-7%	26%	15%	17%
China	16%	-5%	1%	7%	19%
Europe	57%	10%	-3%	7%	13%
India	57%	6%	-3%	7%	13%

To show the total by region, it can stay in the context of the matrix or be plotted into bar chart format. The bar chart emphasizes that India and Europe grew at the same rate.

Revenue Trends



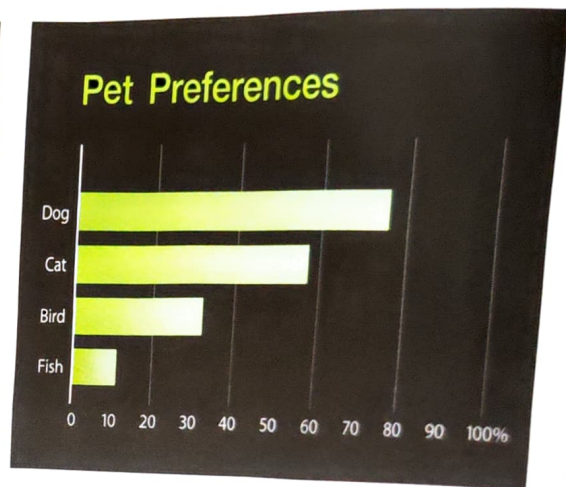
Picking the Right Tool for the Job

Now that you know which results you want to display, you'll need to choose the most appropriate tool for displaying your data. The most frequently used graphs for business communications are pie, bar, and line graphs.

Sometimes though, the best chart is no chart at all. If you only have one key message for your slide or the big message is just a number, why bother putting it in a chart? You could possibly increase the impact of your slide by eliminating the chart altogether. But which slide is more effective at comparing pet preferences?

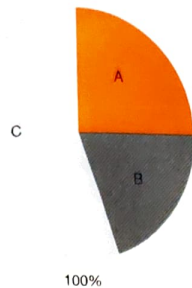


The left slide is effective at communicating the conclusion of the data. The slide on the right shows a comparison of the data.

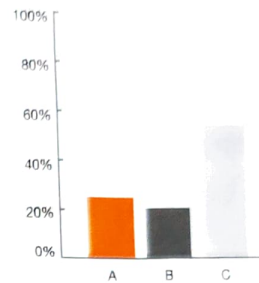


T!P

When several slides in a row have charts, line up the axes of the charts from slide to slide to avoid a jump that jumps around.



Pie charts work only for showing large differences in proportion, especially percentages. Use them when you want to show all of the parts that make up a whole, or compare the percentages of one set to the percentages of another.



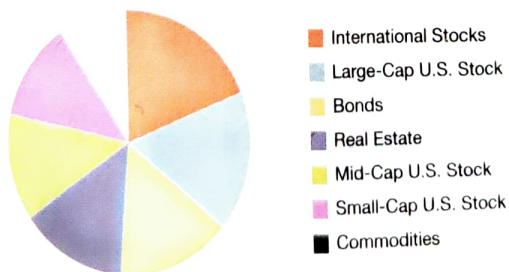
Bar charts are visually more precise than pie charts, and can accommodate larger data sets. Plus, you can stack them to add an additional set of data. Use them when you need to show precise relationships.

TIP

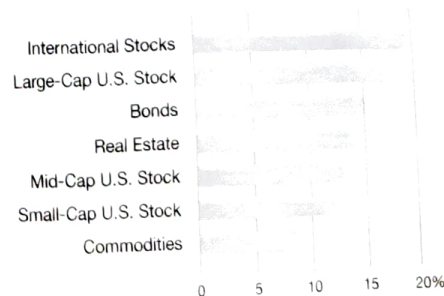
- Always start your first data set for a pie chart at the 12 o'clock position.
- Limit a pie chart to eight sections. More is too many to differentiate on a slide.
- Percentages on a pie chart must add up to 100%.

Which graph below should you use if you wanted the audience to determine whether Mid-Cap U.S. Stock or Small-Cap U.S. Stock has the greatest share?

Images Courtesy of Stephen Few



Investment Portfolio Breakdown



Investment Portfolio Breakdown

Highlighting What's Important

Design rules for showing data on slides differ from the rules of displaying data on paper or through other media. Keep in mind that the purpose of slides is not to show all the data, but to communicate conclusions and insights. Slides can reference your handouts for the dense data. Think of a data chart in three layers.

Background

The background layer contains elements like tick marks, scales, legends, and grid lines. It provides context, scale, and reference for the data. Backgrounds should employ neutral colors.

Data

The data plotted on the slide is usually pulled from a data table. Data can be expressed through various chart types. Whatever the type, a chart should provide enough contrast to allow the audience to clearly distinguish information at a distance without straining.

Emphasis

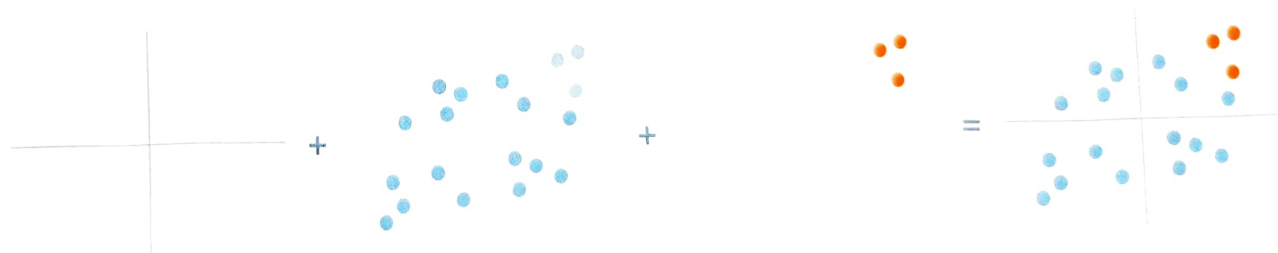
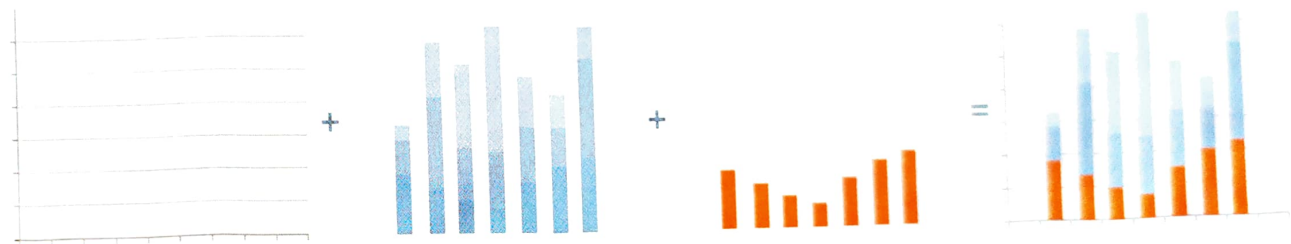
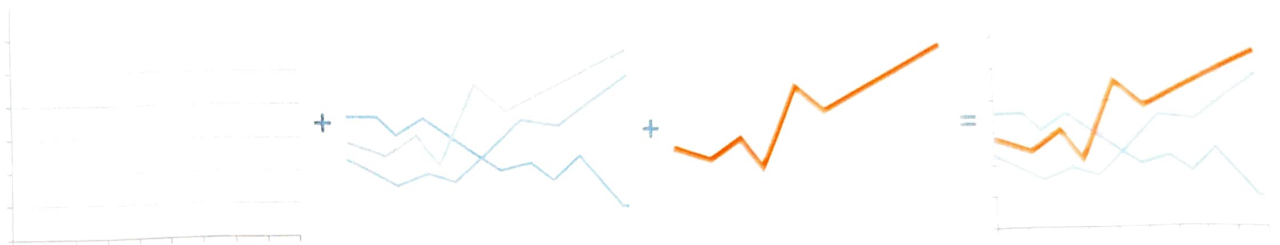
The emphasis layer exists in the foreground and clearly highlights the key message from the data that is most important to your overall message for the slide. This concept might be a data point that's emphasized or a conclusion from the data.

Background

Data

Emphasis

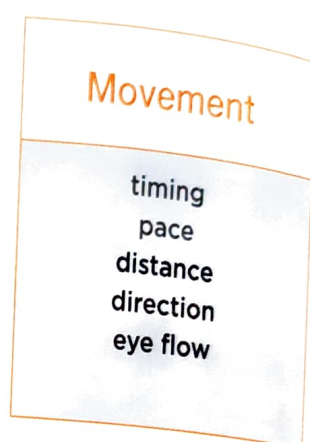
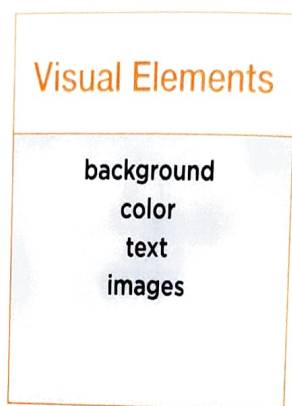
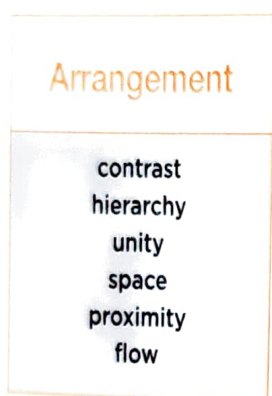
Result



The final charts contain all the data but use contrast to draw the eye to the information that's most critical to the message.

Designing Effective Slides

Approaching the design of a slide can seem daunting. Effective slide design hinges on mastery of three things: arrangement, visual elements, and movement. These will be covered in the following chapters:



Presentations broadcast information to an audience in much the same way as a radio transmission. Thus, think of a presentation's slides as a signal. Can you remember when your favorite junior high song came on the radio but there was so much static interference that you turned off the radio rather than hear your fond memories garbled? Strength and clarity determine how well information is conveyed.

Like a poor radio signal, slides are susceptible to interference and noise, clouding the intended message and compromising the audience's ability to discern message and intent.

The arrangement, visual elements, and movement of a slide function either as a signal, delivering information clearly and directly, or as noise, interfering with the message and causing the viewer to tune out.

In the vast majority of cases, a slide fails not due to stylistic issues, but due to its fundamental construction. Thus, the most important tool in a presenter's arsenal is a strong familiarity with the mechanics of a well-designed slide.

Placement of Elements Creates Meaning

Quite possibly, how slides are arranged has the most impact on whether a slide's message is visually clear. Arrangement tells a story. Based on the arrangement decisions a designer makes, a slide can prompt feelings of tension, confusion, and agitation; conversely, it can maximize clarity by employing the following:

Contrast

The audience can identify the main point quickly.

Flow

The audience knows the order in which to process the information.

Hierarchy

The audience sees the relationship between elements.

Unity

The audience senses that the information belongs together.

Proximity

The audience perceives meaning from the location of elements.

Whitespace

The audience has visual breathing room.

These are the tools of the slide design trade. It's tough to assemble a great slide without paying close attention to each of these issues. Luckily, these are intuitive concepts that may already be familiar to you.

Slides begin and end with ideas. It's your job to take these invisible, abstract ideas and determine how they can be best represented in a tangible, visual form. It's the presenter's responsibility to ensure that the audience is guided through this invisible world by making ideas easy to decode.

Many presenters don't understand arrangement as a principle. They will put everything on one slide and assign them equal value. It is critical to determine which visual elements should have prominence so they attract attention first.

Create dominance with some elements and practice restraint with others. Force yourself to make a decision about the priority of the information.

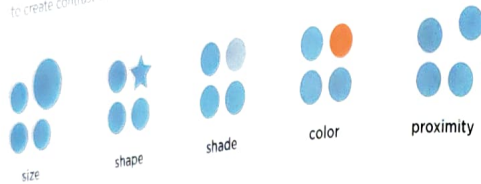
It's laziness on the presenter's part to put everything on one slide.



Overcrowding the slide doesn't add to the clarity of the message.

Contrast: Identifying the Main Point Quickly

Audiences need contrast. Viewers immediately perceive the difference between the attributes of two or more things, and this focuses their attention. There are many ways to create contrast on a slide; here are some examples:



Establishing a relationship between slide elements is an important first step. The contrast in the size of the following text, for example, draws the eye to the larger text as if it's a title or context for the text block. Regardless of where the text is placed, the larger text is perceived as the most important.

Lorem Ipsum

Putpat, conset to od
tut lum nit aufpatem
diamcommy

Lorem Ipsum

Putpat, conset to od
tut lum nit aufpatem
diamcommy

San velenibh ex. Et
vent wisl ulla feugue

Lorem Ipsum

tionsequis nostrud.

Creating contrast in a body of text can be done with size, but also through the use of color. Some versions of the Bible highlight the words of Jesus in red. Below, a quote from Edward Tufte, author of *Visual Explanations*, has a key phrase highlighted in red.

Tufte's design strategy of the smallest effective difference uses contrast wisely yet subtly and only when necessary. He says, "...the idea is to use just notable differences, visual elements that make a clear difference but no more— contrasts that are definitive, effective and minimal."

A common mistake presenters make is assigning contrast unintentionally. Virtually any stylistic difference between two elements suggests something to the viewer on either a conscious or subconscious level. Unintentional contrast can confuse the intended message at best, and contradict it at worst. Remember, all stylistic choices have the potential to suggest importance, urgency, and value. As a result, you should base all such choices on a well-defined purpose.

"Without contrast you're dead."

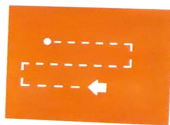
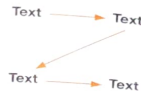
Paul Rand
designer

Flow: Ordering How the Information Is Processed

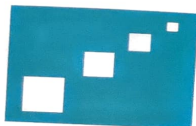
A typical Western reading pattern runs from left to right and top to bottom. Readers are conditioned to start at the top left and scan back and forth across content in a Z-shaped path until they've processed the information.

Jerry Weissman, author of *Presenting to Win*, calls this the conditioned carriage return, as it mimics the movement of the carriage on an old fashioned typewriter.

Readers move their eyes back and forth across a slide until they feel they have identified everything on the slide. They then will assign meaning to the information. If, to make your point, your graphic needs to flow in a direction that's counter-intuitive to natural eye movement, build it over time, with discrete elements appearing in the order you want your audience to process it. Alternately, use a symbol or arrow that clearly marks a starting point. The audience should be able to understand the intended order in which to process the information, without ever feeling lost or overwhelmed.



Using an arrow to mark the starting point draws attention to it where to begin.

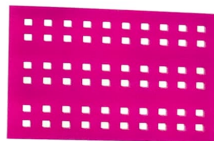


Size and perspective indicate the bottom left as the starting point here.

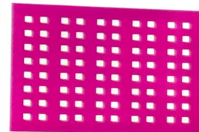
You should organize slides to guide the audience's eyes through the content in an obvious way. This principle applies not only to text, but to diagrams and images as well.



When using images of people, make sure that they're looking at the content instead of looking away, or fleeing it.



At a glance, the audience should be able to determine whether to read data charts horizontally or vertically.



TIP

- Avoid more than three layers of information on a single plane.
- Create points of interest (one main point and up to two sub-points).
- Develop flow within the slide intentionally.
- Choose images and diagrams with clear directional flow.
- Select images that flow toward the focal point on the slide or toward the next slide.

Hierarchy: Seeing Relationships Between Elements

An audience processes hierarchy almost as quickly as contrast. Visual hierarchy, simply put, defines the structure formed when relationships are applied to a set of elements. A simple example of this is the relationship of a title to its body text. Within the hierarchy, the title is the parent and the text is the child. Just like a family lineage, each element in a hierarchy is the child of the element above it, or the parent of the element below it.



Size and location are used together to indicate a decreasing significance from the top left to the bottom right.

Making the bullet points heavier than the title disrupts the logical flow. The audience would have a hard time making sense of it.

Not coincidentally then, the visual structure precisely mirrors the structure of the information it represents. For example, the following diagrams radiate from a core. The core is what binds the elements together and establishes the parent-child relationship. Modifying the size and proximity of the objects changes the meaning behind the relationships and tells a story about their relative importance.

Changing the Size and Proximity of Objects Modifies Their Visual Story



Elements Are Equal
Story: We all need to rally together as a team around this goal for which you're all equally suited.



Parent Dominates
Story: We all need to rally together as a team around this goal which is more important than activities at the team or individual level.



Child Dominates
Story: We all need to rally together as a team around this goal and some of the teams will have bigger challenges than others.

When your presentation is over, you want the audience to walk away understanding their mission. If you don't intentionally provide that message, the audience will create meaning based on their own interpretation of the information. Make sure that you've presented the graphical hierarchy of your message accurately.

The Ingredients of a Great Slide

Now that you know how to arrange elements on a slide, it's time to identify the elements you'll work with. This section covers best practices to use when determining the elements. There really aren't many decisions to make about them; that's why selecting them thoughtfully and carefully is important. The four visual elements are background, color, text, and images. The first three are covered in this section, and images are covered in the next chapter.



Background



Color



Text



Images

Each ingredient—background, color, text, and images—determines how your slide elements will look and requires that you make key design decisions. Once you've decided how the elements will look, you can apply the principles from Chapter 6, "Arranging Elements."

There's one specific theme you need to keep at the forefront of your mind—consistency. Choose element styles and then stick with them. These elements actually become visual triggers of your brand. Whether your brand is just you or a major corporation, pick elements that you feel will resonate with your audience, and then stay consistent.

Still, every blue moon, there is also power in breaking the consistency. Let's say your slide design remains consistent for 20 slides. Then, to create a memorable visual moment, you incorporate something inconsistent with the rest of the slides. That inconsistency will stand out. However, if you break the rules all the time, you've lost your opportunity for visual emphasis.

Background

A background is a container or surface on which to place visual elements. It can incorporate anything you want, or it can have nothing on it at all.

You determine whether the surface is opaque or textured, and whether it has a light source, and from where it originates. But first you need to pry yourself away from the default templates with their preordained slide junk. Think through what is really required. What reflects your intent and personality? What reflects your company's brand? What will act in service to your information rather than compete with it? Consider approaching the background in a way never seen before.



Stockbyte/Getty Images

Where does it say that every slide needs a logo? The people who have come to hear you speak most likely know who you work for. Use your first and last slides (called bumper slides) to identify yourself or company.

On the other hand, if your boss insists you put the logo on every slide, the lower right is the best place for it because you can wrap the right rag of text around it. After all, the same boss who wants the logo on every slide probably has so much text to shoehorn in, you need to wrap it.

Two of my pet peeves are folks who include their logo on every slide, and that animated paper clip in the Microsoft Office product suite.

Two of my pet peeves are folks who include their logo on every slide, and that animated paper clip in the Microsoft Office product suite.



Backgrounds Are a Surface for Digital Assets

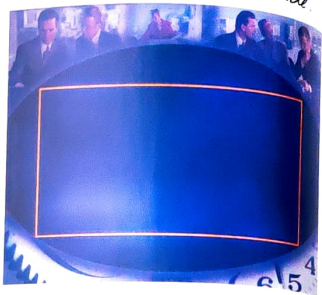
Backgrounds are intended as a surface on which to place elements. They are not in themselves a work of art.

They are a setting, surface, or platform. In print design, designers start with a white or blank piece of paper and then place objects on it. You rarely see a brochure where every page has been covered with a crazy design like the one here.

A background creates a sense of space. That space should be open, spacious, and simple. A slide background is like real estate; it's very valuable, so build on it wisely. Avoid the trend to make the background ornamented, crowded, and distracting. Keep the background as simple and intentionally clean as possible. Your background is there to host your objects; not be an object.

Backgrounds should never compete with content.

The space inside the box is all that's usable.



This is an ineffectual slide template with useless ornamentation around the edges. The orange box highlights the active or "live" area of the slide that can be used for graphics. The designer of this template used at least one-third of the available real estate for meaningless graphical elements.

So now that you've seen a case for open space, what do you do with this nice large open canvas?

Ignore all the default masters and look through your company's marketing material to identify visual elements of the brand that are timeless. You can usually identify a grid system, line structures, bounding boxes, color palette, or visual elements that anchor the designs. Pick the more timeless elements that will never change. You can start there as a basis for the background design. If you don't work for a company, then the sky is the limit. However, your decisions still need to reflect your personality.

Color

Color is crucial to your presentation. It sets a tone and helps establish what the audience will expect. It helps communicate what type of journey you will be taking them on. Is it an exciting journey or a serious or dangerous one? Pick colors that properly represent you, your company, and your stance.

Who is your audience?

Determine colors that appeal to the target audience and avoid colors that won't resonate with them.

What industry are you in?

Review the colors of your competitors to ensure that your chosen palette helps you stand out in your field. Also avoid colors that are negative in your industry. For example, red should be avoided for a financial institution, but would be okay for a blood bank.

Who are you?

Pick colors that reflect your personality. Whether your brand is vivid and exciting or repetitive and conservative, select colors that represent it.

Our family went on a train ride from San Jose to Chicago. Amtrak times it perfectly so that the train arrives in Denver right at sunset. This leg was to be the highlight of the journey—descending the Rocky Mountains into the Colorado basin. But unbeknownst to us, during the night the pass over the Rockies was blocked, and in Salt Lake City we were rerouted through southern Wyoming.

The hope of an entire day meandering through the Rockies and descending into Denver was shattered. Instead of a vivid, vibrant, and exciting palette, we had a consistent, monotonous, and conservative palette. Not that a conservative palette is bad. It just didn't match our expectations. Try to meet your audience's expectations.



The Colorado skies were supposed to be the highlight of our journey



Unfortunately miles of bland landscape deflated the mood of the trip.

About the Color Wheel

Understanding and using the color wheel helps you choose a harmonious palette.

The color wheel helps you visualize the relationships that colors have to one another. Microsoft's PowerPoint and Apple's Keynote applications base their color wheels on the one that Sir Isaac Newton discovered. The wheel uses three primary colors, red, yellow and blue, spaced evenly apart. Blending those colors creates the full color wheel, as shown on the right-hand page.

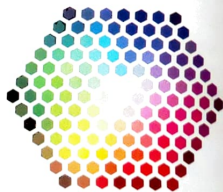
Each pie slice of the wheel has tints and shades of a single hue (true color). The hue on this wheel is four rings out from the center. The colors toward the center of the wheel have white added to the hue, called *tints*, whereas the rings of colors on the outside of the hue have black added to them to create a *shade* of the color.

Presentation applications reference the color wheel's logic in their color picker tools, which is a good reason to understand the color wheel itself.

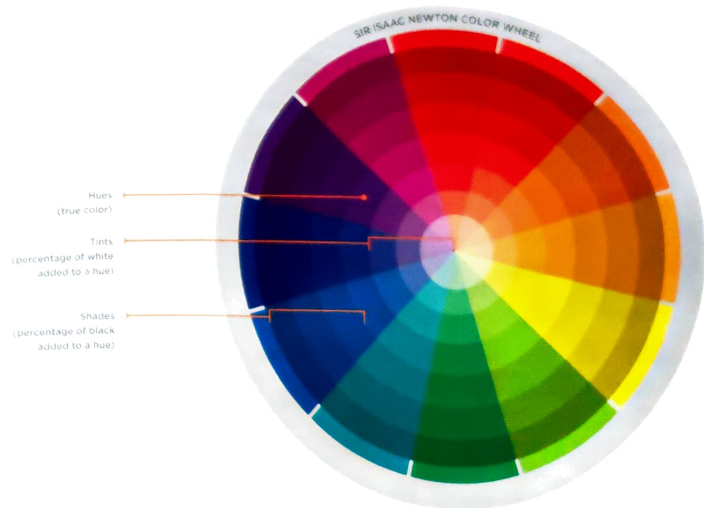
You can select colors for your palette anywhere in the wheel with any saturation of color, but make sure that it contrasts and projects well. There is a bit of a science to creating a pleasing palette.



Apple's Keynote Color Picker



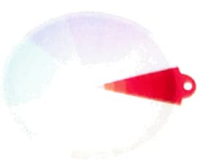
Microsoft's PowerPoint Color Picker



Using the Power of Color

Monochromatic

Using variations of the same color can create a striking palette, but it's difficult to use color effectively without adding black, white, and shades of gray for contrast. Select colors with a broad range of tints and shades to provide enough contrast.



Analogous

Selecting colors that are touching in the wheel creates a narrow, harmonious color scheme. Similar to monochromatic colors, analogous colors provide a predominantly warm or cool feeling depending on where they lie on the wheel.



Complementary

Colors from the opposite ends of the wheel provide the most contrast. If you can shave off the temptation to add more colors, limit them to only two to create a powerful (or even startling) look. But arrange them wisely so they don't clash.



Split Complementary

This variation of the complementary scheme uses two colors on either side of a directly complementary color. These colors have high visual contrast but with less visual tension than purely complementary colors.



Triadic

Three colors equally spaced around the color wheel create vivid visual interest. Some of the palettes are bold while others are more refined. You can modify this method by using a light tint or dark shade of one of the three colors instead of the pure hue.



Tetradic

This scheme is popular because it offers strong visual contrast while retaining harmony. It uses two pairs of complementary colors. It's difficult to harmonize this scheme if all four hues are used in equal amounts. Pick a dominant color for your design and use the others to support the main one.



Earthy

R 72	R 101	R 144
G 107	G 141	G 193
B 28	B 43	B 62

Strong

R 144	R 215	R 243
G 25	G 32	G 113
B 28	B 39	B 84

Powerful

R 136	R 196	R 198
G 20	G 22	G 58
B 119	B 28	B 31

Calm

R 144	R 215	R 243
G 25	G 32	G 113
B 28	B 39	B 84

Athletic

R 202	R 0
G 108	G 84
B 24	B 150

Feminine

R 191	R 188
G 86	G 211
B 139	B 87

Regal

R 84	R 206	R 92
G 39	G 138	G 160
B 133	B 20	B 56

Retro

R 47	R 241	R 246
G 179	G 86	G 150
B 202	B 79	B 84

Inviting

R 0	R 123	R 206
G 124	G 10	G 128
B 128	B 107	B 20

Playful

R 13	R 208	R 252
G 124	G 63	G 238
B 193	B 65	B 33

Spirited

R 0	R 27	R 138	R 202
G 82	G 125	G 13	G 103
B 143	B 55	B 16	B 32

Healthy

R 103	R 144	R 199	R 192
G 163	G 193	G 77	G 104
B 189	B 62	B 31	B 138

Choosing Your Colors

Before you can decide on a color palette, you need to determine your background color. With the technological advancement of projectors, there are no restrictions on what background color to use anymore, because most colors now translate well. However, in certain circumstances, a darker background might be more effective than a lighter background, and vice versa.

Two factors determine whether dark or light is appropriate: the formality of the event and the venue size. The human eye requires contrast for visibility, and pure black or pure white backgrounds have the greatest opportunity for contrast since they are without color. Look at the color palettes on the right. Here the PowerPoint color palette appears on various colored backgrounds. The palettes on the black and white backgrounds on the opposite page are completely visible and have full contrast. On the midrange colored backgrounds, the contrast is diminished. Some of the colors are indistinguishable from the background, and portions of the palette are unusable.

When choosing your color palette, make sure that it contrasts with the background and the other colors you have chosen—and holds up to a projection test. Colors might look great on your computer screen, but then look different or diluted when projected. If color integrity is important to you, either use your own projector as often as possible or arrive at your venue early enough to have time to adjust the projector.

Dark background

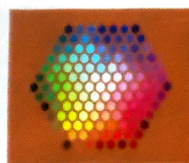
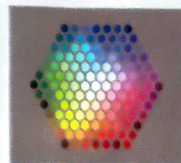
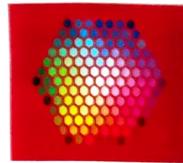
- Formal
- Doesn't influence ambient lighting
- Does not work well for handouts
- Fewer opportunity for shadows
- For large venues
- Objects can glow



Palettes and light gradients can be "blown out" by the projector and appear indistinguishable from each other by the audience.

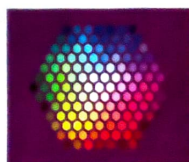
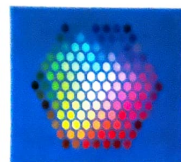
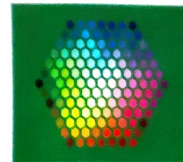
Light background

- Informal
- Has a bright feeling
- Illuminates the room
- Works well for handouts
- For smaller venues (conference rooms)
- No opportunity for dramatic lighting or spotlights on the elements



Contrasting with the background is critical so the audience can see your content.

In these samples, some colors disappear, but many blend in. That's because the background is a midtone instead of very dark or very light. Midtones do not favor the contrast.



Text

Remember the 3-second rule.

Presentations are a "glance media"—more closely related to billboards than other media. It requires commuters to process quickly as they drive past. Imagine having a billboard full of bullets; drivers would crash trying to process the ads.

Ask yourself whether your message can be processed effectively within three seconds. The audience should be able to quickly ascertain the meaning before turning their attention back to the presenter. For comparison, consider that when billboards first appeared, a public outcry ensued over concerns of driver safety and visual pollution of the landscape. Interestingly, relatively few protests have aired against the visual pollution in meeting rooms across corporate America.

Content should lend itself to quick processing, and should be typeset correctly.

When Apple's Macintosh first debuted, anyone with a computer could now be a desktop publisher. At Duarte, we dove in head first and were thrilled when a reputable design firm invited us in for a portfolio review. While reviewing our work, the Creative Director's expressions and line of questions made me squirm. Over and over, she asked probing questions about our typesetting decisions. I could tell she was not impressed. She told us that we needed to master type if we wanted to stand out from all the novices entering the field in droves. We took her advice and studied the masters.



Her advice was timely. Shortly after, we were approached by Adobe and Apple, who both treat type as a critical component to the success of their brands. They took as much care around the treatment of type as they did their concept, copy, images, and layouts. Our accurately typeset presentations helped us land these accounts.

Some might say that typesetting isn't worth the extra time for a presentation. But type is a fundamental part of our culture and has been for centuries. Most people can inherently and easily recognize when type is balanced and used well.

Before computers, typesetting was a specialized skill and honorable profession. Digitization of fonts and the advent of the personal computer spawned a massive generation of users who don't understand the context and beauty in carefully typeset design.

Dissecting a Font

Every font has its own personality: serious or playful, stately or friendly, bold or humble. Here are two different fonts. Study them and look for the obvious and subtle differences. When you've trained yourself to see the differences, you can make better choices about which font is the most appropriate for your presentation.

Consider the height and weight. Look at the various shapes created by the ascenders and descenders. Notice the unique shapes formed in the negative space of the letters, called *counters*. The two words shown here are the same point size, but the x-height is taller in the serif font. Choose a font that accurately reflects the personality of your brand—like the examples shown to the right.



don't vs. don't

There isn't much that turns my stomach more than seeing someone use a foot mark where an apostrophe should be. Make sure your apostrophes have a head and a tail. If they look like splinters it's bad; if they look like tadpoles, it's good.

Serif

Serifs are intended for a long sequence of words that exceed one line. They have little feet that guide the letters into each other so the letters appear connected, and to help the eye stay within the line of text in dense copy. Serifs possess varying line weights that help the eye identify the letter quickly.

Sans Serif

The word "sans" means without, so sans serif means without serifs. The letterforms are bigger and bolder. Sans serif typefaces are usually used in children's books because of their simplicity. Some believe they are more difficult to read, so they are usually used in short bursts like headlines, subtitles, and captions.

The debates still rage about which type is most suited for legibility. The results of many studies are inconclusive, so I conducted my own research from the 28th floor of a Las Vegas hotel. If my earlier premise is true that slides are most comparable to billboards, then sans serif is the way to go. Out of the 40 or so billboards visible from the hotel room, the only ones I could read were set in sans serif type. Debate over, I guess.

A fundamental element of design is how the fonts are combined, scaled, tabulated, and spaced.

Font

Personality

Georgia	formal, practical
Times New Roman	professional, traditional
Courier	plain, nerdy

Arial	stable, conformist
Tahoma	young, plain
Century Gothic	happy, elegant

Just like great wine pairing, font pairing can amplify great design. It is best to combine no more than two fonts per presentation. Use one font for headlines and subheads, and the other for blocks of text.

If you need a special font for emphasis, that might justify a third, but rarely. Instead, use color or italics to create emphasis. More than two fonts will make your slides look busy. Choose your primary typeface wisely and select a secondary font to complement it. It's also perfectly fine to use only one font throughout your presentation.

How Many Words Should Be on a Slide?

There really are no official rules on the word count for a slide. Ultimately, you need enough words to make you comfortable delivering your message. Put enough there to serve as a mnemonic, but go for a very low word count. If you use a plethora of words, your audience will read the slide more quickly than you can explain it, making you strangely irrelevant to your own presentation. One time, for fun, I delivered a presentation by just being quiet and letting my staff read the slides. It was a hoot. We were done at least ten minutes sooner!

The default template in PowerPoint is a slideument. This is not a visual aid; it is a document. Avoid two-line titles when giving a presentation because of the distance the eye has to travel across the slide. In fact, consider doing a presentation with titles only, like the one below.

Big idea here

Simple Works!

- Succinct text
- Crisp thoughts
- Big ideas
- Clear mnemonic
- Relieved audience

The Default Blank Template Accommodates a Two-line Title

- Primary bullet is 32 points, which is good
 - Then return and tab for the second level
- Notice that the bullet style changes at each level
 - With each new tab the font's point size is reduced
 - The smallest text is 20 points, which is slideument size
- When you move to a second point
 - It becomes obvious that this just might be
 - Something you want the audience to read
 - Instead of hearing you present the content
 - In compelling and human way

The standard default PowerPoint template above encourages two-line titles and sub, sub, sub, sub, sub points. It's a document, not a slide.

Revealing Text

Now that you've determined how much text is on a slide, you need to decide in what order and how quickly the audience should see the text. I prefer to have text build sequentially as I'm not sure why anyone would want the audience to jump ahead. Remember, if the audience can see your bullets, they know the points you're going to make. They'll get bored or agitated waiting for you to catch up with them.

If you choose to repeat an agenda slide in your presentation to give the audience context, gray out previous text so they can visually jump to the current point.

TIP

Don't animate your text unless it adds value, meaning, or emotion to the content.

Agenda

- Agenda Item #1
- **Agenda Item #2**
- Agenda Item #3

Contrasting Text

"Grayed" Out Text

Contrasting Text

"Grayed" Out Text

Contrasting Text

"Grayed" Out Text

Use a lighter or darker shade of the background color to dim text.

Typesetting a Block of Text

Paragraph spacing determines the gap between blocks of text. In the case of bullet points, each bullet is considered a new paragraph, whereas a second line in a bullet is not.

Can you see that just above this line of text, the space is larger than the type within the paragraph? Line spacing is the space between lines within a paragraph. Holding down the Shift key and then pressing Return (called a soft return) creates line spacing, whereas pressing the Return key alone (called a hard return) will give you paragraph spacing.

This is paragraph spacing!

Title placed here

- Insert your bullet line
- Insert your bullet line
- Insert your bullet line
- Insert your bullet line

Bullets

Notice the paragraph spacing between each bullet paragraph.

Title placed here

- Major Bullet number 1
 - Sub-bullet number 1
 - Sub-bullet number 2
- Major Bullet number 2
 - Sub-bullet number 1
 - Sub-bullet number 2

Sub-bullets

If you insist on using them, they should be spatially associated with the primary bullet. Notice how there's a larger space between the sub-bullet and next major bullet. That's paragraph spacing.

In typesetting terms, a widow is a single word by itself at the end of a paragraph (in this case it is the word "performs").

"The typographer must analyze and reveal the inner order of the text, as a musician must reveal the inner order of the music he performs."

Robert Bringhurst

The above text is aligned left. The rag is the shape that the text makes on the nonaligned side (in this case the right side). Look at the shape and determine if it has a distracting shape. In this case the word "reveal" sticks out too much, and the word "performs" in the last line is a widow. Usually all can be fixed easily by making a minor adjustment to the width of the text block, making manual line breaks with a soft return, or both.

"The typographer must analyze and reveal the inner order of the text, as a musician must reveal the inner order of the music he performs."

Robert Bringhurst

Adjusting the text box creates a straighter rag and removes the widow.

Validating Your Font Size

What is the most appropriate minimum font size? If you have to ask, you may be using your slides as a document. Here are a few good approaches:



- 1 Measure the diagonal length of your computer screen. Let's say it's a 21 inch monitor. Using a tape measure, place a piece of tape 21 feet from your screen. If your screen is 17 inches, place the tape 17 feet away, etc. Then, launch a slide on your screen into slide show mode. Whatever you cannot see from behind the piece of tape probably can't be seen by the back of the room.
- 2 Put your file into slide sorter view. Look at the slides at 66 percent size. If you can still read them, so can your audience.
- 3 Stand in the back of the room at your venue and click through all the slides so you know what people in the back row will see.
- 4 Follow the advice of Guy Kawasaki, author and former Apple Fellow: "A good rule of thumb for font size is to divide the oldest investor's age by two, and use that font size."

TIP

Stay consistent. For keynotes, don't go smaller than 28 pt. If you are consistently reducing your point size to under 24 pt and using third-level bullets, you have officially created a document and not a slide.

Planning Animations

According to the book of Ecclesiastes, "There's a time to live and a time to die." The same is true for animations. There is a time for them to live, and there's a time for them to just NOT!

If you choose to use animation, it should look natural and alive.

The word animation is derived from the Latin word *anima*, which loosely means "the breath of life." Most presentation animation sucks the life out of the audience instead of breathing life into it.

The movement of objects should seem familiar and make sense.

Animation should help you understand the interrelatedness or sequence of information you're presenting. The audience should be able to process the information in the order that the animation presents it.

Because of the important role motion plays in a story, Hollywood painstakingly plans every scene and movement. Storyboards detail each scene change and each move the actor makes is extensively planned.

In presentations, the actors are your elements, and each needs to serve in helping tell your story better. You may not need to create elaborate storyboards, but you should identify the places where animation will help get your point across, and invest planning, thinking, and time into making it powerful.

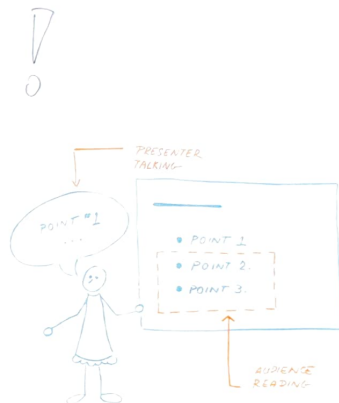
Make sure that you're synchronized with your slides. By nature, viewers will read and process the visual information the moment it's presented. This can create confusion when a slide has several visuals, but the speaker is expounding on only one. The audience will try to listen while reading ahead at the same time.

Hide elements until you need to refer to them. This ensures the item being discussed will remain the focus of your audience—resulting in the audience listening more and reading less. Which is the point of having a speaker in the first place.

TIP

Slow moving animation (the Ken Burns type) can help create a feeling of nostalgia or even the passage of time.

Fast-moving animation (quick cuts) can help create a sense of excitement, energy, or surprise.



Avoiding Visual Vertigo

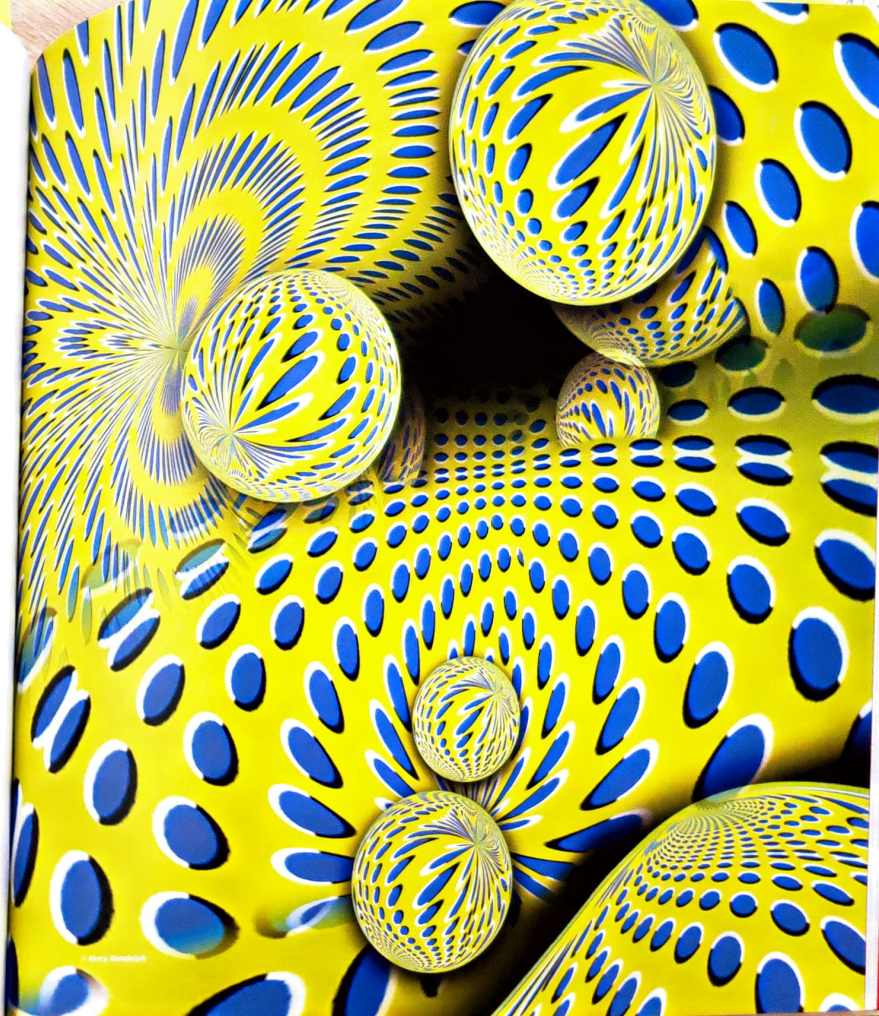
I recently attended a presentation at a swanky executive briefing center in India. They'd invested big bucks in an animated corporate overview showcased in a plush room that looked like the bridge of the Starship Enterprise. Of the 20 executives in the room, two of them became so motion sick that they sat through most of the presentation with their eyes closed. That's ten percent of the audience lost due to an overzealous animator.

Just because an animation feature is in an application doesn't mean you have to use it.

For many of the companies Duarte works for, we will ban specific animations from their use because they are counter-cultural to the brand personality of that company. It's tough to find a good use for many of the custom animation features in PowerPoint. You need to pretend they aren't there.

You Know You're Using Animation Wrong When

- 1 The animation is unnatural or counterintuitive.
- 2 The pace is frenetic, annoying, or chronic (buzzes like a fly).
- 3 It doesn't add value to the content or serve a purpose.
- 4 The movement distracts the audience from the presenter.
- 5 The animation feature is used "just because."
- 6 Too many animations confuse the purpose.
- 7 The animation style is inappropriate for the content.
- 8 A sudden animation surprises or startles the audience.



Making Template Design Decisions

Build a template that is timeless so that you won't tire of it easily. If built well, it can remain in circulation for years.

When a large company transitions to a new template, the process can take a Herculean effort that tanks productivity, at least temporarily. The best way to avoid having to re-create the template every time there's an inspiring new ad campaign or marketing whim is to focus on the timeless components of your brand or industry. Those elements are the visual attributes of the brand that rarely, if ever, change. At right are the brand attributes that evolve versus those that remain relatively unchanged. Don't make design decisions that will expire in the short term, or base your decisions on a short-term campaign.

Look through all the material that your company produces and identify the elements that are purely decorative versus those that are the truly essential visual brand elements. For the template, use the essential elements only.

Remember, the template is just that, a template. It should be a basic shell for your words and images. It shouldn't be a stand-alone work of art.

Rarely Change

- 1 Logo
- 2 Signage
- 3 Template-driven systems
 - Business systems (letterhead)
 - Website framework
 - Brochure templates
 - Datasheets
- 4 Visual attributes
 - Color palettes
 - Grid layouts
 - Fonts
 - Graphical elements

Often Transform

- 1 Ad campaigns
- 2 Marketing campaigns
- 3 Retail displays and packaging
- 4 Annual reports

You need to determine what should go in a template, compared to what should go on the slide. To do that, you'll need to identify the varying types of content that your company uses to convey its messages. One approach is to collect files from many departments, audit them, determine communication patterns, and then build a template that meets the communication needs of each organization appropriately.

The effective use of a template helps viewers become accustomed to where the content will appear on a slide and give them a sense that slide content is structured and aligned. Content should not bounce from slide to slide. There's nothing more frustrating than having content jump around so much that each slide becomes a puzzle that takes some study to figure out where to start processing information. Titles and bullet copy should appear in the same locations (unless you've intentionally designed it that way to serve a purpose). By setting the placement and divisions of space for your corporate power-users, they will be able to recognize (and value) the underlying skeletal structure, the reasoning used to develop the template, and the rationale for them to use the template.

visual style

TIP

Believe it or not, before you start developing a template, a conversation with your IT department is in order.

- Ask the IT department what the migration plan is for the company's presentation application. If the company is close to a major upgrade, get that beta and begin a strategy on how to build a template in the newest release of the application.
- If you create a template loaded with images, query the IT department whether the organization has adequate storage. Presentation files procreate like bunnies, and a 1MB template might seem innocent, but it can add up to terabytes in a single department in a month.

The Power of Constraints

There's not much on this planet more gratifying than being in a healthy and comfortable relationship. My husband and I often use the metaphor of a steamboat to describe our interaction: I'm the one above the surface, sputtering, churning, and making noise to create propulsion; he's the quiet rudder under the surface providing guidance and wisdom to make sure that we don't run aground. I would describe our relationship as interdependent. We rely on and respect what the other brings to the partnership, and that's what makes it successful.

But how would you characterize the relationship you have with your visual aids? What metaphor would you use to describe it? Actor and stage? Or drug and drug user?

When you present, are you codependent, interdependent, or even independent of your slides?

The term codependent was coined in the mid '80s and initially used by Alcoholics Anonymous to describe the unhealthy psychological condition that occurs in a relationship when one partner suffers from an addiction.

A codependent person ends up providing too much care—and often too much latitude—to someone who is unhealthy. You are in a codependent relationship with your slides if you are desperately addicted to them and unable to break away. You may even make excuses for their sad condition, which of course only perpetuates their dilapidated state. If you are in a work culture that enables this codependent relationship, it will be even tougher to transform yourself.

The first step is to admit you have a problem. The second is to acknowledge that your slides have become unmanageable.

Dense content on slides can be addictive. The default template settings in applications such as PowerPoint encourage you to put an enormous amount of text on each slide, which transforms the slide into a document rather than a visual aid. The slides take on the role of teleprompter, requiring you to read them to the audience instead of connecting with your listeners. This habit requires little or no effort on the part of presenters to rehearse; they simply need to read aloud along with the audience. Thus, instead of becoming true "carriers" of your content, you may be happy merely to repeat your own pre-written words. Though definitely the easiest and quickest way to present—it requires no effort on your part other than writing the presentation—is such an approach really best for your audience? It's frustrating to sit through a presentation, reading bullet points, waiting for the presenter to catch up while expanding on the already very thorough and dense text. The audience can read, you know. PSHAW.

Using this medium improperly creates the impression that the presentation is solely about the slides and the presenter, and not about the audience's comprehension or insights. Originally, slides were intended as a visual aid for the audience—a mnemonic device to help them retain and recall the material being presented. Slides are like the stage in a play or the costume for the actor; they are not themselves the star of the show, but they do provide the context and setting in which you can give a memorable performance.

So what can you do if you're trapped in a codependent relationship with your slides that negatively impacts your audience? Garr Reynolds of *Presentation Zen* says that the concept of "restrictions as liberators" can be applied to most any presentation situation. By constraining and restricting yourself and your media, you can break your codependent relationship. But it will take hard work and commitment.

Over the next few pages you'll learn how to constrain your text, the length of your presentation and the output of your projector to help benefit the audience.



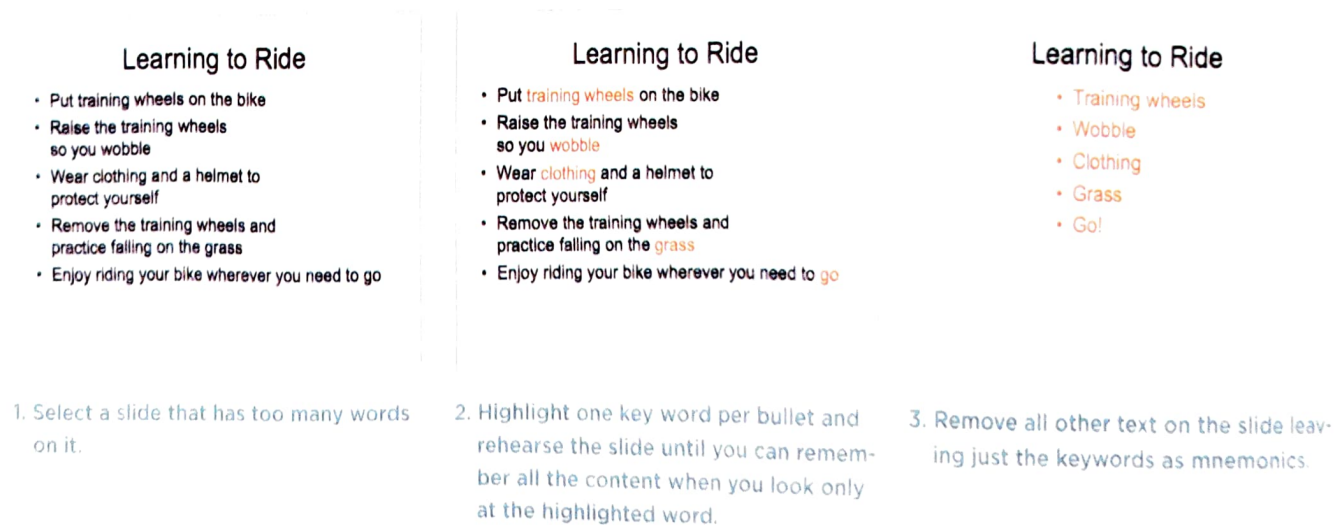
Reducing Text on a Slide

Remember how scary it was to learn to ride a bike? The first step was getting your mind and body to understand the mechanics of the handlebars. It felt awkward having your feet push the pedals in a circle to turn the wheels that would propel you forward. Getting to this part, though, was relatively easy because you could learn those skills within the safety of the training wheels. It wasn't until you felt comfortable with these new sensations that you either had someone take the training wheels off or bend them up a bit so you could become familiar with wobbling and the concept of self-balancing. You were keenly aware of the risk of falling and getting hurt.

This same fear of failure traps us into not wanting to let go of our heavy text. Letting go is a process, and it takes practice. And yes, possibly falling down. But the practice has a liberating payoff for you and the audience.

Great presenters connect with their audience, speak naturally, and allow the slides to enhance their story.

But that's easier said than done! Memorizing an entire presentation may seem difficult and time-consuming. But it's not as hard as you might think. Here's an exercise to help wean you from the content on your slides.



Constraining the Length

Have you ever finished a presentation and had folks flock to you begging you to make it longer next time? Probably not. Attention spans are getting shorter and shorter. Thirty-minute sitcoms, ten-minute YouTube videos, and 15-second commercials all influence the duration of our attention span. And that short span takes into account that TV and Internet media are usually professionally prepared and most often geared to entertain. So, when your content isn't emotionally stimulating but is still crucial and necessary to effective communication, you can enhance the impact by packing punch in a smaller time slot. Dragging the audience down Dullard Lane isn't anyone's favorite afternoon trip.

Constraining your presentation's length forces you to be concise and remove anything superfluous to the message. Proof in point is the recent release of TED videos online (www.ted.com). TED (Technology, Entertainment, Design), an annual conference in California featuring some of the greatest thought leaders and innovators in the world, helps spread ideas and shape our culture and future. We see how each of these phenomenal presenters communicates his or her ideas in 18 minutes or less. So, if some of the most influential people in the world can deliver powerful content in 18 minutes, can you, too. Just try boiling it down.

Another popular presentation technique that uses time as a constraint is Pecha Kucha (see page 228 and www.pechakucha.org). Inaugurated in Tokyo by Klein Dytham architecture and various collaborators, Pecha Kucha brings together designers and thinkers from various industries for an evening of presentations. Each presenter exhibits only 20 slides, each automatically advancing after 20 seconds, for a total of six minutes and 40 seconds of ideas and inspiration.

For internal company meetings, constraining presentations and reserving some of the meeting for conversation, collaboration, and Q&A would be refreshing. Some companies are beginning to constrain the number of slides allowed and encouraging a rapid-fire approach so that more insights can be shared in a single setting.

"Don't you hate it when the presenter has crammed so much information into a presentation that there's no possible way they can get through it all without talking like an auctioneer? They avoid questions and discussion because 'we've got a lot to get through.'"

No, we don't.

Presenters bring it upon themselves by trying to cover *Moby Dick*, *War and Peace*, and *Les Misérables* when we only had time for *One Fish, Two Fish, Red Fish, Blue Fish*."

Tom Jamison
Author, www.drathernewriting.com

Just right:

Too many
pages of
material for
one hour:



Constraining the Projector

A further step is to admit you can be PowerPoint-less.

What would it be like to have a relationship independent of slides? Is it possible to deliver a presentation without any slides? Consider what would happen to your presentation if a power outage occurred. Evaluating and rejecting bad habits that have formed from relying on presentation software is critical to your success. It will force you to take ownership of the content in your head and heart.

There are several ways to use visual aids that don't involve a projector. Flip charts, props, handouts, flip books, and even physical devices work if you have a small audience. Using more tactile forms of communication with a small audience increases your humanness. On the other hand, if you are in a large venue with a projector, you can increase your presence by pressing the B key on the keyboard. This will turn your screen to black, forcing your audience to focus on you alternately, the W key will change your screen to white.

Moreover, if your keyboard is out of range, slip an all black slide into your deck. This is equivalent to giving the audience a visual pause.

As you know, if you pause during your presentation, it creates more drama and meaning, and reinforces what you have to say.

Going to black gives the audience time to contemplate. Similarly, turning off the projector during a critical part of your presentation allows the audience to feel as if they are interacting solely with you, which can build credibility and trust.



When in slide show mode, press the B key to turn your screen to black so that focus is on the speaker. Alternatively, press the W key for an all white slide.

How Many Slides? Use the 10/20/30 Rule.

As a venture capitalist, Guy Kawasaki has to listen to hundreds of entrepreneurs pitch their companies. Most of the pitches are crap: sixty slides about a "patent pending," "first mover advantage," and "all we have to do is get one percent of the people in China to buy our product." So he came up with the 10/20/30 rule for PowerPoint. It's quite simple: a PowerPoint presentation should have 10 slides, last no more than 20 minutes, and contain no font smaller than 30 points. He believes that this rule applies to presentations when the audience needs to reach consensus.

Kawasaki says that 10 slides is the optimal number of slides because a normal human being cannot comprehend more than 10 concepts in a meeting—and venture capitalists are very normal. If you must use more than 10 slides to explain your business, you probably don't have a business.

The Ten Topics That a Venture Capitalist Cares About

- | | |
|-------------------------------|-------------------------------|
| 1 Problem | 4 Competition |
| 2 Solution | 7 Team |
| 3 Business model | 8 Projections and milestones |
| 4 Underlying magic/technology | 9 Status and timeline |
| 5 Marketing and sales | 10 Summary and call to action |

You should deliver your 10 slides in 20 minutes. Sure, you have an hour time slot, but you're using a Windows laptop, so it will take 40 minutes to make it work with the projector. Even if setup goes perfectly, people will arrive late and have to leave early. In a perfect world, you give your pitch in 20 minutes, and you have 40 minutes left for discussion.

The majority of the presentations that Guy sees have text in a 10-point font. As much text as possible is jammed into the slide, and then the presenter reads it. However, as soon as the audience figures out that you're reading the text, they read ahead of you. The result is that you and the audience are out of sync.

The reason presenters use a small font is twofold: first, they don't know their material well enough; second, they think that more text is more convincing. Total bozosity. Force yourself to use a point size no smaller than 30 points. It will make your presentations better because it requires you to find the most salient points and to know how to explain them well.

This content was slightly modified from Guy's blog at www.guykawasaki.com

Kawasaki's approach challenges the traditional venture capitalist approach of "walk softly and carry a big deck." This one-size-fits all approach might not work in every situation. VCs being persuaded to part with millions of dollars want to have some content in writing to sink their teeth into and pass around in e-mails. Details and facts are important to them, too—just as important as the clarity of message.

So it's recommended that if you use the 10/20/30 rule of your presentation application, if you design your slides and the Notes master layout well, these handouts can be almost as sophisticated as a brochure.

On the next four pages, you'll see other approaches to slide count that work well under different circumstances.



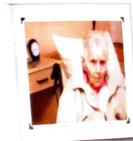
And that's when Dr. Thomas discovered the problem with nursing homes—(LIFE) (MUSIC)

He wasn't able to find a cure for Grapes' sneezes anywhere in his black medical bag.

He couldn't find a prescription drug that would help.

And none of his medical texts even MENTIONED the word sneeziness.

In the end, he had to walk out of Grapes' room without being able to ease her suffering, and he returned to his office, determined to find a cure for sneeziness.



Then Dr. Thomas' colleagues' were alerted to his other patients.

When he told his next "meeting" (MUSIC) that they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC).

Even when "Dr. Thomas" told his colleagues that "Dr. Thomas" was "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC).

When he told his colleagues that "Dr. Thomas" was "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC) then they were all "not sneezing" (MUSIC).

Lawrence is the only doctor of his kind.